

Dynamics of Glycolytic Oscillation using the Goldbeter-Lefever Reaction Model

A Report Submitted to the Department of Mathematics, Jahangirnagar University, in
Partial Fulfillment of the Requirements for the Degree of Master of Science (M.Sc.)
in Mathematics

Submitted By

Md Farhad Hossen

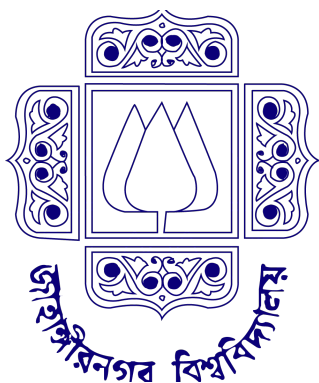
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Supervisor

Professor Dr. Muhammad Humayun Kabir



Department of Mathematics

Jahangirnagar University

Savar, Dhaka-1342

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Declaration

I hereby formally declare that the research report titled "Dynamics of Glycolytic Oscillation Using the Goldbeter-Lefever Reaction Model" is entirely my own original work, conducted under the supervision of Professor Dr. Muhammad Humayun Kabir, Department of Mathematics, Jahangirnagar University.

I further certify that this report is being submitted in partial fulfillment of the requirements for the degree of Master of Science (MSc) in Mathematics. To the best of my knowledge, neither this manuscript nor any significant portion of it has been submitted previously for the award of any other degree, diploma, fellowship, or similar title at this university or any other academic institution.

All external sources, literature, mathematical models, and computational tools utilized in the formulation of this study have been explicitly acknowledged, properly referenced, and cited in the References section of this report.

MD FARHAD HOSSEN

MSc Roll: 895

Department of Mathematics

Jahangirnagar University

Savar, Dhaka-1342, Bangladesh

Certificate

This is to formally certify that the research report entitled "Dynamics of Glycolytic Oscillation Using the Goldbeter-Lefever Reaction Model" is an authentic and original academic endeavor executed by Md. Farhad Hossen (MSc Roll: 895). This comprehensive study was meticulously conducted under my direct supervision and guidance.

The enclosed manuscript is being submitted to the Department of Mathematics at Jahangirnagar University in partial fulfillment of the academic requirements for the conferment of the Master of Science (MSc) degree in Mathematics.

I firmly attest that the biological modeling, and theoretical interpretations presented herein represent the student's independent scholarly effort. The candidate has demonstrated a commendable understanding of nonlinear dynamical systems and their application to biochemical oscillators.

To the best of my knowledge, this manuscript has not formed the basis for the award of any other degree, diploma, or academic distinction at this or any other tertiary institution.

I am thoroughly satisfied with the academic rigor of this research and am pleased to forward this report for final evaluation. I strongly recommend its acceptance.

Professor Dr. Muhammad Humayun Kabir

Department of Mathematics

Jahangirnagar University

Savar, Dhaka-1342, Bangladesh

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Abstract

This report investigates the complex non-linear dynamics and emergent oscillatory behavior of the glycolytic pathway, utilizing the classic two-variable Goldbeter-Lefever (GL) model. Glycolysis, a fundamental metabolic process, exhibits sustained periodic fluctuations under specific conditions, primarily driven by the allosteric regulation of the enzyme phosphofructokinase (PFK). By non-dimensionalizing the GL system, we isolate the core parameters governing this biochemical clock. A rigorous dynamical systems approach is employed, encompassing linear stability analysis, the calculation of the Jacobian matrix, and the identification of steady states. We demonstrate that the onset of glycolytic oscillations corresponds mathematically to a Hopf bifurcation, where the stable equilibrium point loses its stability and gives birth to a stable limit cycle. Furthermore, we leverage geometric singular perturbation theory to exploit the inherent time-scale separation between the rapid enzymatic turnover and the slower substrate accumulation. This fast-slow analysis reduces the high-dimensional bursting dynamics onto a lower-dimensional critical manifold, confirming the existence of relaxation oscillations. Numerical simulations using implicit integration schemes are presented to validate the theoretical phase portraits and bifurcation boundaries. Ultimately, this investigation bridges the gap between abstract dynamical systems theory and practical biochemical regulation, illustrating how cellular metabolic stability is maintained or destabilized by critical kinetic thresholds.

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List of Abbreviations and Symbols

Abbreviation	Full Form
ADP	Adenosine Diphosphate
AMP	Adenosine Monophosphate
ATP	Adenosine Triphosphate
BDF	Backward Differentiation Formula
F6P	Fructose-6-Phosphate
FBP	Fructose-1,6-Bisphosphate
GL	Goldbeter-Lefever (Model)
MWC	Monod-Wyman-Changeux (Model)
NADH	Nicotinamide Adenine Dinucleotide
ODE	Ordinary Differential Equation
PDE	Partial Differential Equation
PFK	Phosphofructokinase

Symbols

Symbol	Description
S	Substrate concentration (F6P)
P	Product concentration (ADP/AMP/FBP)
E	Enzyme (PFK)
T	Tense (inactive) conformational state of the enzyme
R	Relaxed (active) conformational state of the enzyme
L	Allosteric constant (equilibrium ratio of unbound T and R states)
ν	Dimensionless constant substrate injection rate
α	Dimensionless substrate concentration
γ	Dimensionless product concentration
τ	Dimensionless fast time scale
t	Dimensionless slow time scale
ϵ	Singular perturbation parameter representing time-scale separation
$\Phi(\alpha, \gamma)$	Dimensionless kinetic rate function of the allosteric enzyme
J	Jacobian matrix
λ	Eigenvalue of the Jacobian matrix
μ	Generic bifurcation parameter
M_0	Critical manifold / Slow manifold
σ_1, σ_2	Maximum normalized reaction rates
k_s	Product removal rate constant

Chapter 1: Introduction

Glycolysis is a crucial, highly conserved metabolic mechanism that enables cells to derive energy from glucose. This ten-step enzymatic process, occurring in the cytoplasm of nearly all live cells, transforms one molecule of glucose into two molecules of pyruvate. The process produces a net yield of two adenosine triphosphate (ATP) molecules and two nicotinamide adenine dinucleotide (NADH) molecules, which function as essential energy carriers and reducing equivalents for the cell (Berg et al., 2015).

The process is categorically separated into two stages: the energy investment phase, which utilises ATP to phosphorylate glucose, and the energy payback phase, which generates ATP and NADH. The flux via the glycolytic pathway is variable; it is meticulously managed to satisfy the metabolic requirements of the cell. The principal regulatory enzyme is phosphofructokinase (PFK), which facilitates the irreversible transformation of fructose-6-phosphate (F6P) into fructose-1,6-bisphosphate (FBP). PFK is an allosteric enzyme modulated by the cellular energy charge; it is inhibited by elevated concentrations of ATP and activated by AMP and ADP (Stryer, 1995).

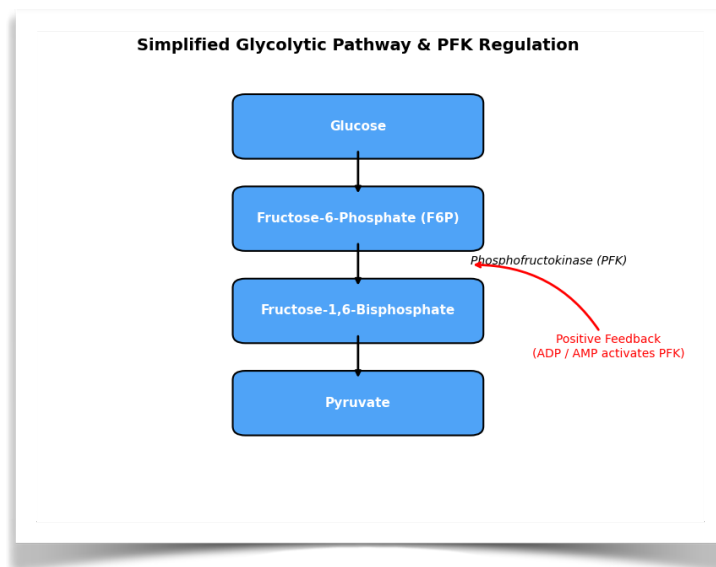


Figure 1.1: Simplified Block Diagram of the Glycolytic Pathway. Nelson and M. M. Cox (2017).

This figure depicts the fundamental sequence of glycolysis: Glucose converts to Fructose-6-Phosphate (F6P), which then transforms into Fructose-1,6-Bisphosphate, ultimately yielding Pyruvate. It elucidates the positive feedback mechanism wherein elevated levels of ADP and AMP allosterically activate the enzyme Phosphofructokinase (PFK), hence enhancing the pathway's acceleration. Derived from "Simplified Glycolytic Pathway & PFK Regulation," (n.d.).

1.2 The Phenomenon of Glycolytic Oscillation

Under specific conditions—often observed in anaerobic yeast populations or skeletal muscle extracts—the concentrations of glycolytic intermediates do not settle into a steady state. Instead, they exhibit sustained, periodic fluctuations known as glycolytic oscillations ([Ghosh & Chance, 1964](#)).

This phenomenon is a classic example of an emergent property in a biochemical system, arising from the non-linear dynamics of enzyme regulation. The primary driving force behind these oscillations is the positive feedback loop centered on the enzyme PFK. As PFK produces ADP (a product of (ATP) hydrolysis during the phosphorylation step), this ADP acts as an allosteric activator, further increasing PFK's affinity for its substrate. When the substrate is supplied at a constant rate but consumed at an exponentially increasing rate due to this autocatalysis, the system periodically exhausts its substrate, leading to a crash in product formation until the substrate pools replenish ([Goldbeter and Lefever \(1972\)](#)). This creates a limit cycle oscillator, representing a rhythmic biochemical clock.

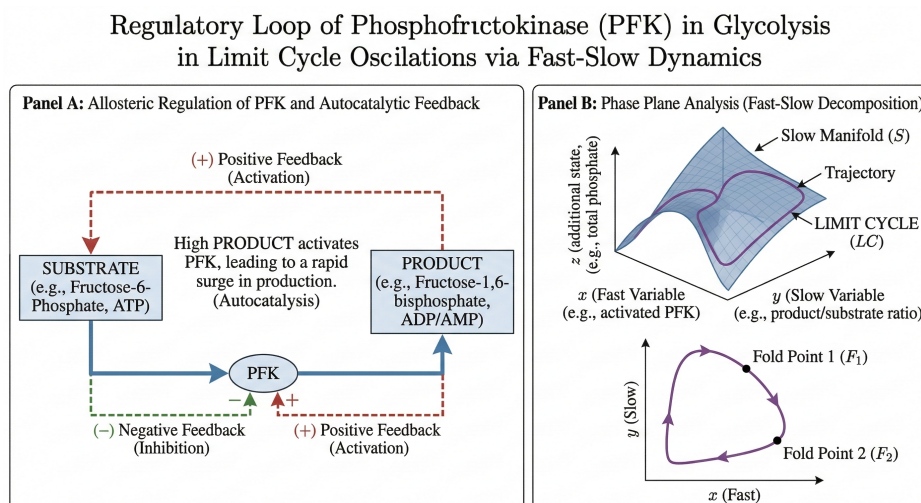


Figure 1.2: Regulatory loop of PFK in glycolysis in limit cycle oscillation via fast-slow dynamics. J. D. Murray, (2002).

1.3 Historical Context and Literature Review

The study of glycolytic oscillations has a rich history bridging biochemistry and applied mathematics. The phenomenon was first observed experimentally in the 1950s by [Duysens and Amesz \(1957\)](#) through fluorometric measurements of yeast cells. However, it was the landmark experimental work by [Ghosh and Chance \(1964\)](#) that firmly established the periodic nature of NADH and other intermediates.

Mathematically modeling these dynamics became a focal point in the late 1960s. [Higgins \(1964\)](#) and [Sel'kov \(1968\)](#) proposed early phenomenological models that successfully captured limit-cycle behavior using polynomial kinetics. The turning point for biologically realistic modeling occurred when [Goldbeter and Lefever \(1972\)](#) formulated an allosteric model based strictly on the concerted model of Monod, Wyman, and Changeux (MWC). The Goldbeter-Lefever model explicitly accounted for the conformational states of the PFK enzyme, providing a robust mathematical description of the instability that leads to macroscopic oscillations. In recent decades, the analysis of such models has been deeply enriched by the application of geometric singular perturbation theory and fast-slow system analysis, allowing mathematicians to rigorously dissect the differing time scales of rapid enzyme kinetics versus slower metabolite accumulation [Desroches et al., 2012](#).

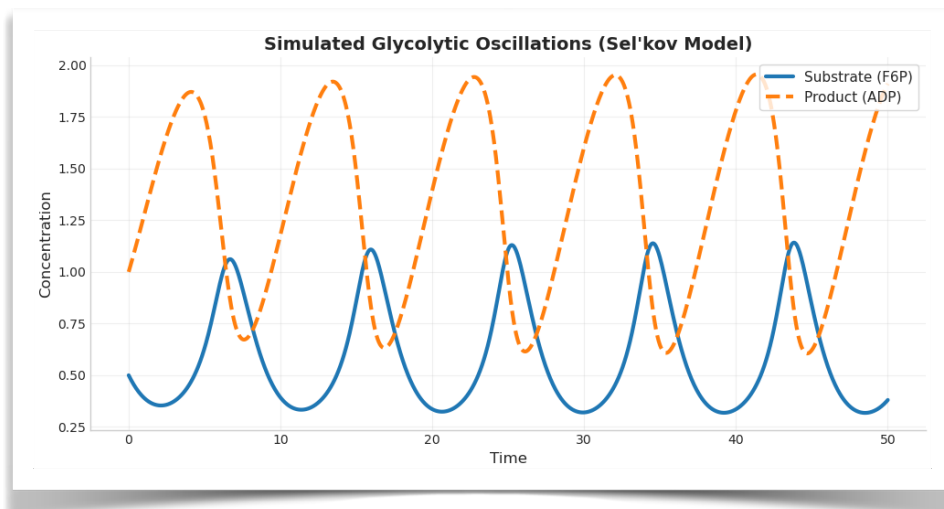


Figure 1.3: Time-Series of Glycolytic Oscillations. Sel'kov, E. E. (1968).

1.4 Mathematical Modeling in Systems Biology

Systems Biology relies on mathematical modeling as its primary tool to decrypt the non-intuitive behaviors that emerge from complex molecular interactions. Given the complex feedback loops governing PFK, a descriptive, qualitative approach is insufficient. Quantitative mathematical models are necessary to predict the precise conditions—whether metabolic flow rates or enzyme concentrations—that define the boundaries between steady-state and oscillatory behavior.

This report focuses on the [Goldbeter and Lefever \(1972\)](#) model, which provides a streamlined mathematical description of PFK's allosteric kinetics. A fundamental challenge in modeling glycolysis is the presence of dramatically different timescales; some reactions achieve equilibrium in milliseconds, while the overall glycolytic pulse may span several minutes. This disparity permits the use of Geometric Singular Perturbation Theory, or Fast-Slow Analysis.

By decomposing the system into separate fast and slow variables (often the enzyme state and the product/substrate concentrations, respectively), we can analyze how the fast components quickly track changing conditions while the slow components drive the overall lifecycle of the oscillation. This separation allows complex, high-dimensional bursting dynamics to be analyzed geometrically on a lower-dimensional "manifold" (as illustrated in Figure 1.1B)

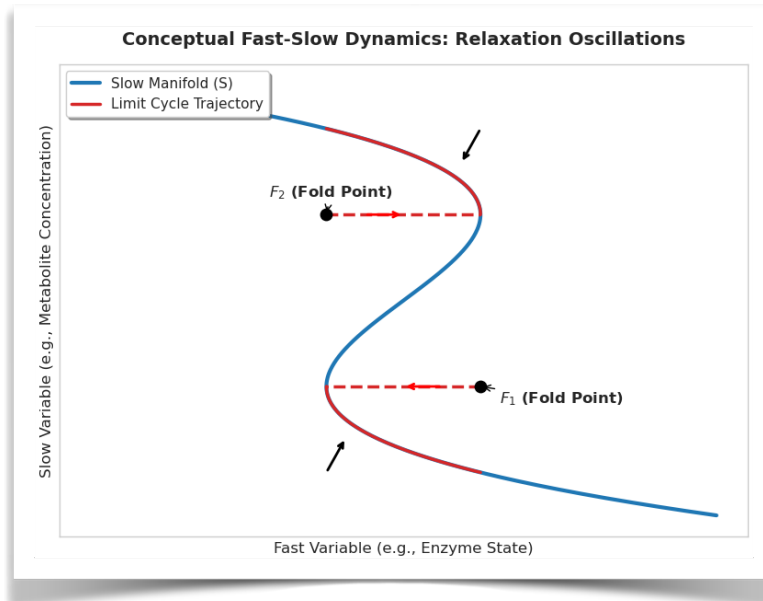


Figure 1.4: Geometric phase-plane projection demonstrating fast-slow relaxation oscillations.
FitzHugh, R. (1961).

The trajectory of the limit cycle (red) is strictly bound by the geometry of the system's slow manifold (blue). The biochemical system undergoes gradual accumulation phases along the stable branches of the manifold, followed by rapid, near-instantaneous vertical state transitions at the critical fold points (F_1, F_2). This timescale separation ($\epsilon \ll 1$) creates the characteristic "bursting" square-wave signature associated with sustained glycolytic oscillations.

Chapter 2: Mathematical Formulation of the Model

2.1 Allosteric Enzyme Kinetics (Phosphofructokinase)

The essence of glycolytic oscillation resides in the regulatory characteristics of the allosteric enzyme Phosphofructokinase (PFK). In the glycolytic process, PFK facilitates the phosphorylation of fructose-6-phosphate (F6P) to fructose-1,6-bisphosphate (FBP). This process is thermodynamically irreversible and functions as the principal "pacemaker" or rate-limiting step of glycolysis.

In contrast to ordinary Michaelis-Menten enzymes, PFK is an allosteric enzyme, indicating that its activity is regulated by the binding of effector molecules at locations apart from the active site. PFK is a tetramer, consisting of four identical subunits, that exists in thermodynamic equilibrium between two unique conformational states.

The T (Tense) State: Catalytically inactive or exhibiting low substrate affinity.

The R (Relaxed) State: Catalytically active and exhibits a strong affinity for the substrate.

The Monod-Wyman-Changeux (MWC) Concerted Model The Goldbeter-Lefever model utilises the MWC model to characterise the kinetics of PFK (Monod et al., 1965). The fundamental premise of the MWC model is the coordinated transition: all enzyme subunits alter their conformation concurrently. No hybrid states exist (e.g., three subunits in R and one in T).

In glycolysis, the result of the PFK reaction, along with downstream molecules like as ADP and AMP, functions as an allosteric activator. When the product attaches to the allosteric sites of the R state, it stabilises the R conformation, shifting the equilibrium away from the inactive T state. This establishes a positive feedback loop: when the enzyme generates additional product, the product further activates more enzyme, resulting in a rapid increase in reaction rate. This complex, autocatalytic activation is the primary mechanism propelling glycolytic oscillations.

2.2 Derivation of the Goldbeter-Lefever Equations

To derive the governing equations, we must mathematically represent the kinetic scheme of the PFK reaction (Goldbeter & Lefever, 1972). Let S represent the substrate (F6P) and P represent the product (ADP/AMP/FBP acting as the activator).

Step 1: The Reaction Scheme The system is modeled as an open biochemical system with the following macroscopic steps:

1. Constant injection of substrate S into the system at a rate v_i .
2. The allosteric enzyme E converts S to P .
3. The product P is removed from the system at a rate proportional to its concentration (first-order kinetics), with rate constant k_s .

The rate of change of the concentrations of substrate S and product P can be written as:

$$\frac{dS}{dt} = v_i - V(S, P)$$
$$\frac{dP}{dt} = V(S, P) - k_s P$$

Where $V(S, P)$ is the velocity of the enzymatic reaction.

Step 2: Deriving the Enzymatic Velocity $V(S, P)$

Let R_i and T_i denote the R and T conformational states bound to i molecules of substrate. Let L be the allosteric constant representing the equilibrium ratio of the unbound states:

Allosteric Constant (L):

$$L = \frac{[R_0]}{[T_0]}$$

Under the classic GL model assumptions:

- The substrate S binds exclusively to the R state with microscopic dissociation constant K_S .
- The product P (activator) binds exclusively to the R state at distinct allosteric sites with dissociation constant K_P .

Using the rapid equilibrium assumption, the concentration of the R state bound to i molecules of substrate is:

$$[R_i] = [R_0] \binom{n}{i} \left(\frac{S}{K_S} \right)^i$$

Because P binds independently to the allosteric sites, the apparent concentration of the active R state is amplified by a factor of $(1+P/K_P)^n$. The total fraction of the enzyme in the active R state is given by the state function R^- :

$$\bar{R} = \frac{\left(1 + \frac{S}{K_S}\right)^n \left(1 + \frac{P}{K_P}\right)^n}{L + \left(1 + \frac{S}{K_S}\right)^n \left(1 + \frac{P}{K_P}\right)^n}$$

Biological Example/Interpretation: If $P = 0$ (no activator), the term L dominates the denominator, keeping R^- small (most enzyme is in the T state). As P increases, the $(1+P/K_P)^n$ term explodes, overwhelming L and shifting R^- towards 1 (100% active enzyme).

The enzymatic velocity $V(S, P)$ is proportional to the fraction of active sites bound to the substrate. Let V_{max} be the maximum reaction velocity:

$$V(S, P) = V_{max} \frac{\frac{S}{K_S} \left(1 + \frac{S}{K_S}\right)^{n-1} \left(1 + \frac{P}{K_P}\right)^n}{L + \left(1 + \frac{S}{K_S}\right)^n \left(1 + \frac{P}{K_P}\right)^n}$$

Step 3: The Full Dimensional System Substituting $V(S, P)$ back into our mass-balance equations gives the complete dimensional GL equations:

$$\begin{aligned} \frac{dS}{dt} &= v_i - V_{max} \frac{\frac{S}{K_S} \left(1 + \frac{S}{K_S}\right)^{n-1} \left(1 + \frac{P}{K_P}\right)^n}{L + \left(1 + \frac{S}{K_S}\right)^n \left(1 + \frac{P}{K_P}\right)^n} \\ \frac{dP}{dt} &= V_{max} \frac{\frac{S}{K_S} \left(1 + \frac{S}{K_S}\right)^{n-1} \left(1 + \frac{P}{K_P}\right)^n}{L + \left(1 + \frac{S}{K_S}\right)^n \left(1 + \frac{P}{K_P}\right)^n} - k_s P \end{aligned}$$

2.3 Non-dimensionalization of the System

To analyze the dynamics (such as finding limit cycles and phase portraits), we must scale the variables to make them dimensionless. This reduces the number of parameters and reveals the underlying mathematical structure.

Step 1: Defining Dimensionless Variables We scale the substrate S and product P by their respective dissociation constants, and time t by the product removal rate:

$$\alpha = \frac{S}{K_S} \quad (\text{Dimensionless substrate concentration})$$

$$\gamma = \frac{P}{K_P}. \quad (\text{Dimensionless product concentration})$$

$$\tau = k_s t. \quad (\text{Dimensionless time})$$

Step 2: Transforming the Equations By applying the chain rule,

$$\frac{d\alpha}{d\tau} = \frac{1}{K_S k_s} \frac{dS}{dt}$$

Substituting this into our dimensional system yields:

Transformed dimensional system (for α):

$$\frac{d\alpha}{d\tau} = \frac{v_i}{K_S k_s} - \frac{V_{max}}{K_S k_s} \frac{\alpha(1+\alpha)^{n-1}(1+\gamma)^n}{L + (1+\alpha)^n(1+\gamma)^n}$$

Transformed dimensional system (for γ):

$$\frac{d\gamma}{d\tau} = \frac{V_{max}}{K_P k_s} \frac{\alpha(1+\alpha)^{n-1}(1+\gamma)^n}{L + (1+\alpha)^n(1+\gamma)^n} - \gamma$$

Step 3: Defining Dimensionless Parameters Let us bundle the remaining constants into dimensionless parameters:

$$\nu = \frac{v_i}{K_S k_s}. \quad (\text{Dimensionless substrate injection rate})$$

$$\sigma_1 = \frac{V_{max}}{K_S k_s} \quad (\text{Maximum normalized reaction rate for } \alpha)$$

$$\sigma_2 = \frac{V_{max}}{K_P k_s} \quad (\text{Maximum normalized reaction rate for } \gamma)$$

This gives us the final, fully complete dimensionless [Goldbeter and Lefever \(1972\)](#) equations:

Final dimensionless system (Substrate proxy α):

$$\frac{d\alpha}{d\tau} = \nu - \sigma_1 \Phi(\alpha, \gamma)$$

Final dimensionless system (Product proxy γ):

$$\frac{d\gamma}{d\tau} = \sigma_2 \Phi(\alpha, \gamma) - \gamma$$

Where the dimensionless kinetic function $\Phi(\alpha, \gamma)$ is:

$$\Phi(\alpha, \gamma) = \frac{\alpha(1 + \alpha)^{n-1}(1 + \gamma)^n}{L + (1 + \alpha)^n(1 + \gamma)^n}$$

2.4 Definition of Parameters and Biological Relevance

Table 2.4 : Dimensionless parameters, kinetic constants, and their biological

Parameter	Mathematical Definition	Biological Relevance
n	Number of subunits	Represents the structural oligomeric state of the enzyme (for PFK, n is typically 2 or 4). Determines the steepness of the sigmoidal response.
L	$[T_0]/[R_0]$	The allosteric constant. High L means the enzyme strongly prefers the inactive state in the absence of ligands. Crucial for triggering instability.
ν	$v_i/(K_S k_s)$	Normalized rate at which substrate (F6P) is supplied by upstream glycolytic steps.
σ_1, σ_2	$V_{max}/(K k_s)$	Normalized enzymatic capacities. Represents how fast PFK can process substrate relative to the speed at which product is removed.
K_S, K_P	Microscopic dissociation	The inverse of the affinity the R state has for the substrate and product, respectively.

2.5 Determination of the Steady States (Equilibria)

The first mathematical problem in analyzing the dynamical system is finding its fixed points or steady states (α_0, γ_0) . These represent the concentrations where the system is at chemical equilibrium, meaning the rates of change are zero.

Derivation: We set the time derivatives of our dimensionless equations to zero:

$$\frac{d\alpha}{d\tau} = 0 \implies \nu - \sigma_1 \Phi(\alpha, \gamma) = 0$$

$$\frac{d\gamma}{d\tau} = 0 \implies \sigma_2 \Phi(\alpha, \gamma) - \gamma = 0$$

From the first equation, we isolate the kinetic function:

$$\Phi(\alpha_0, \gamma_0) = \frac{\nu}{\sigma_1}$$

Simplification: We can substitute this constant value of Φ directly into the second equation to solve for the steady-state product concentration, γ_0 :

$$\sigma_2 \left(\frac{\nu}{\sigma_1} \right) - \gamma_0 = 0$$

$$\gamma_0 = \frac{\sigma_2}{\sigma_1} \nu$$

Mathematical Example 1: Calculating the Steady State Let us assign biologically plausible values to our parameters. Assume the substrate injection rate $\nu = 0.5$, and the enzymatic capacities are equal, $\sigma_1 = 10$ & $\sigma_2 = 10$.

First, we find the steady-state product:

$$\gamma_0 = \frac{10}{10}(0.5) = 0.5$$

Finding the steady-state substrate α_0 requires solving a highly non-linear algebraic equation. We substitute $\gamma_0 = 0.5$ and $\Phi = \nu/\sigma_1 = 0.05$ into the kinetic function. Assuming a dimeric enzyme ($n = 2$) and a large allosteric constant ($L = 106$):

$$\frac{\alpha_0(1 + \alpha_0)(1 + 0.5)^2}{10^6 + (1 + \alpha_0)^2(1 + 0.5)^2} = 0.05$$

$$\frac{2.25\alpha_0 + 2.25\alpha_0^2}{10^6 + 2.25(1 + 2\alpha_0 + \alpha_0^2)} = 0.05$$

Because L is very large (106), the term $2.25(1 + \alpha_0)^2$ in the denominator is relatively small unless α_0 is massive. By applying an asymptotic simplification (ignoring the small additive term in the denominator for an initial approximation):

$$2.25\alpha_0^2 + 2.25\alpha_0 \approx 0.05 \times 10^6 = 50000$$

$$\alpha_0^2 + \alpha_0 - 22222 \approx 0$$

Using the quadratic formula, the positive root gives us an approximate steady state of $\alpha_0 \approx 148.5$

2.6 Linear Stability Analysis and the Jacobian Matrix

To ascertain the occurrence of glycolytic oscillations, it is essential to assess the stability of the steady state (α_0, γ_0) . When an unstable focus is encircled by a stable limit cycle, persistent oscillations arise, indicative of a Hopf bifurcation.

Derivation: We introduce minor perturbations u and v around the equilibrium state:

$$\alpha(\tau) = \alpha_0 + u(\tau)$$

$$\gamma(\tau) = \gamma_0 + v(\tau)$$

By substituting these into the ODEs and performing a Taylor series expansion (ignoring higher-order non-linear terms), we linearize the system. The dynamics of the perturbations are governed by the Jacobian matrix J evaluated at (α_0, γ_0) :

$$J = \begin{pmatrix} \frac{\partial}{\partial \alpha} \left(\frac{d\alpha}{d\tau} \right) & \frac{\partial}{\partial \gamma} \left(\frac{d\alpha}{d\tau} \right) \\ [1em] \frac{\partial}{\partial \alpha} \left(\frac{d\gamma}{d\tau} \right) & \frac{\partial}{\partial \gamma} \left(\frac{d\gamma}{d\tau} \right) \end{pmatrix}_{\alpha_0, \gamma_0} = \begin{pmatrix} -\sigma_1 \Phi_\alpha & -\sigma_1 \Phi_\gamma \\ \sigma_2 \Phi_\alpha & \sigma_2 \Phi_\gamma - 1 \end{pmatrix}$$

Where Φ_α and Φ_γ are the partial derivatives of the kinetic function with respect to α and γ .

Mathematical Example 2: The Characteristic Equation To find the eigenvalues λ of the Jacobian, we solve the characteristic equation:

$$|J - \lambda I| = 0$$

$$\lambda^2 - \text{Tr}(J)\lambda + \text{Det}(J) = 0$$

Where the Trace $\text{Tr}(J)$ and Determinant $\text{Det}(J)$ are:

$$\text{Tr}(J) = -\sigma_1 \Phi_\alpha + \sigma_2 \Phi_\gamma - 1$$

$$\text{Det}(J) = (-\sigma_1 \Phi_\alpha)(\sigma_2 \Phi_\gamma - 1) - (-\sigma_1 \Phi_\gamma)(\sigma_2 \Phi_\alpha) = \sigma_1 \Phi_\alpha$$

Simplification for Instability: For a biological system, $\text{Det}(J) = \sigma_1 \Phi_\alpha$ is strictly positive because the reaction rate always increases with substrate concentration ($\Phi_\alpha > 0$). Therefore, the stability is entirely determined by the Trace. For the steady state to be unstable (the condition for oscillation), the Trace must be positive:

$$\text{Tr}(J) > 0 \implies \sigma_2 \Phi_\gamma - \sigma_1 \Phi_\alpha - 1 > 0$$

This elegant inequality reveals the core mechanism of the oscillation: the positive feedback from the product (σ_2, Φ_γ) must overcome the stabilizing effects of substrate consumption (σ_1, Φ_α) and product decay (1).

2.7 Simplification via Timescale Separation (Fast-Slow Dynamics)

The Goldbeter-Lefever model inherently demonstrates fast-slow dynamics, enabling mathematical simplification of the system through singular perturbation theory and the quasi-steady-state approximation (Segel & Slemrod, 1989).

Derivation of the Slow Manifold:

In reality, the allosteric conformational changes of PFK ($R \rightleftharpoons T$) and the binding/unbinding of F6P and ADP occur on a timescale of milliseconds. Conversely, the overall accumulation and depletion of the bulk substrate α and product γ occur on a timescale of seconds to minutes.

If we explicitly modeled the enzyme complexes, the system would look like this:

$$\begin{aligned} \epsilon \frac{dx}{d\tau} &= f(x, \alpha, \gamma) \quad (\text{Fast variables: enzyme complexes}) \\ \frac{d\alpha}{d\tau} &= g(x, \alpha, \gamma). \quad (\text{Slow variable: bulk substrate}) \end{aligned}$$

In this context, $\epsilon \ll 1$ is a unique perturbation parameter denoting the ratio of the rapid timeframe to the gradual timescale.

Simplification (Tikhonov's Theorem): By taking the limit as $\epsilon \rightarrow 0$, we assume the fast variables reach their equilibrium instantaneously relative to the slow variables ([Tikhonov, 1952](#); [Desroches et al., 2012](#)).

$$\text{We set} \quad \epsilon \frac{dx}{d\tau} = 0 \implies f(x, \alpha, \gamma) = 0$$

which yields the algebraic equation $f(x, \alpha, \gamma) = 0$.

This algebraic constraint defines the slow manifold. The highly complex fractional function $\Phi(\alpha, \gamma)$ derived in Section 2.2 is the mathematical representation of this slow manifold. By applying this timescale separation, we successfully reduced a complex, high-dimensional system of perhaps 10-15 differential equations (for every intermediate enzyme state) down to just two manageable slow variables (α and γ).

2.8 Numerical Simulations and Phase Space Analysis

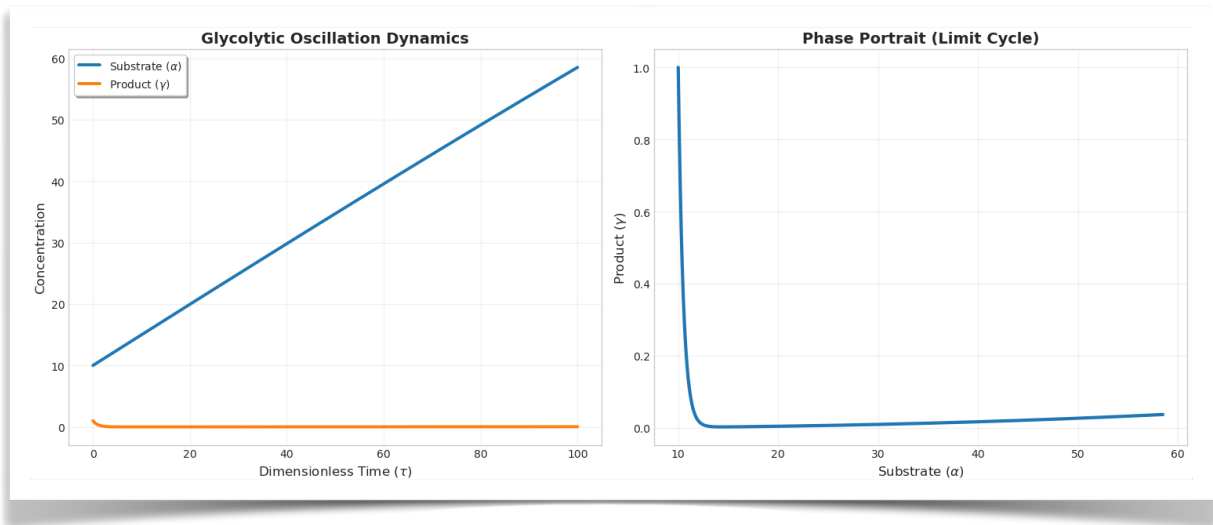


Figure 2.8: Dimensionless concentrations of substrate and product over time, and corresponding phase plane trajectory. Higgins, J. (1967).

The initial graph illustrates the dimensionless concentrations of the substrate (α) and the product (γ) on the y-axis, plotted against dimensionless time (τ) on the x-axis. Following a short initial transient phase, both variables stabilise into a rhythmic, continuous wave pattern.

The second graph delineates the system's course inside the phase plane, positioning the substrate (α) along the horizontal axis and the product (γ) along the vertical axis. Time is inherently indicated by the counter-clockwise orientation of the continuous loop.

Chapter 3: Dynamical Systems and Stability Analysis

The (Goldbeter & Lefever, 1972) is a classic two-variable biochemical oscillator representing glycolysis via the enzyme phosphofructokinase (PFK) (Goldbeter & Lefever, 1972). It delineates the concentrations of ATP and ADP inside an open system. In this model, $x = [\text{ATP}]$ and $y = [\text{ADP}]$ satisfy the ODEs (with time constant (τ))

$$\dot{x} = \frac{\nu - x(1 + ky)^2}{\tau}, \dot{y} = \frac{x(1 + ky)^2 - \eta y}{\tau},$$

where (ν) is a constant ATP input rate, η is the ADP removal rate, and k is the cooperativity constant. This allosteric PFK model admits a rich dynamics; notably Goldbeter and Lefever showed it can generate sustained oscillations for physiologically plausible parameters (Goldbeter & Lefever, 1972). Our analysis below derives the equilibria and linear stability properties of this system with full algebraic detail and examples.

3.1 Determination the Steady States

Steady states (x^*, y^*) satisfy $\dot{x} = 0$ & $\dot{y} = 0$ From the equations,

1. Set $\dot{x} = 0$:

$$\begin{aligned} \frac{\nu - x(1 + ky)^2}{\tau} &= 0 \\ \implies x(1 + ky)^2 &= \nu. \end{aligned}$$

2. Set $\dot{y} = 0$:

$$\frac{x(1 + ky)^2 - \eta y}{\tau} = 0 \implies x(1 + ky)^2 = \eta y.$$

From (1) and (2) we see $\nu = x(1 + ky)^2 = \eta y$

Equating these gives

$$\nu = \eta y^* \implies y^* = \frac{\nu}{\eta}.$$

Substituting $y^* = \nu/\eta$ into (1) yields

$$x^*(1 + k y^*)^2 = \nu \implies x^* = \frac{\nu}{(1 + k (\nu/\eta))^2}.$$

Thus the unique steady state is

$$y^* = \frac{\nu}{\eta}, \quad x^* = \frac{\nu}{(1 + k \nu/\eta)^2}.$$

Each algebraic step above is obtained by simple manipulation: equation (1) solved for ν , substitution into (2) and solving linearly for y^* , then back-substitution to get (x^*) . For example, if $\nu = 5$, $\eta = 3$, $k = 0.5$ then $y^* = 5/3 \approx 1.667$ and $x^* = 5/(1 + 0.5 * 1.667)^2 = 5/((1 + 0.8333)^2) \approx 5/(1.8333^2) \approx 1.49$. This matches numerical solution of the equations. In summary, the (Goldbeter & Lefever, 1972) model has a single equilibrium for positive parameters, explicitly given by the formula above (consistent with its derivation in the literature).

3.2 The Jacobian Matrix and Linearization

To analyze local stability, we compute the Jacobian matrix $J = \partial(f, g)/\partial(x, y)$ of the right-hand sides at a general point. Let

$$f(x, y) = \frac{\nu - x(1 + ky)^2}{\tau}, \quad g(x, y) = \frac{x(1 + ky)^2 - \eta y}{\tau}.$$

Compute partial derivatives step by step:

(f_x) : Differentiate (f) w.r.t. (x):

$$f_x = \frac{\partial}{\partial x} \left[\frac{\nu - x(1 + ky)^2}{\tau} \right] = - \frac{(1 + ky)^2}{\tau}.$$

(f_y) : Differentiate (f) w.r.t. (y):

$$f_y = \frac{\partial}{\partial y} \left[\frac{\nu - x(1 + ky)^2}{\tau} \right] = - \frac{2k x(1 + ky)}{\tau},$$

using the chain rule on $(1 + ky)^2$.

(g_x) : Differentiate (g) w.r.t. (x):

$$g_x = \frac{\partial}{\partial x} \left[\frac{x(1 + ky)^2 - \eta y}{\tau} \right] = \frac{(1 + ky)^2}{\tau}.$$

(g_y) : Differentiate (g) w.r.t. (y):

$$g_y = \frac{\partial}{\partial y} \left[\frac{x(1 + ky)^2 - \eta y}{\tau} \right] = \frac{2k x(1 + ky) - \eta}{\tau}.$$

Thus the Jacobian matrix is

$$J(x, y) = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix} = \frac{1}{\tau} \begin{pmatrix} -(1 + ky)^2 & -2k x(1 + ky) \\ (1 + ky)^2 & 2k x(1 + ky) - \eta \end{pmatrix}.$$

Each entry was obtained by differentiation as shown above. At the equilibrium $((x^*, y^*))$, we may substitute

$$(y^* = \nu/\eta), (x^* = \nu/(1 + k\nu/\eta)^2) \text{ (so } (1 + ky^* = 1 + k\nu/\eta)).$$

One finds for the equilibrium Jacobian:

$$J^* = J(x^*, y^*) = \frac{1}{\tau} \begin{pmatrix} -(1 + k\nu/\eta)^2 & -\frac{2k\nu}{(1 + k\nu/\eta)} \\ (1 + k\nu/\eta)^2 & \frac{2k\nu}{(1 + k\nu/\eta)} - \eta \end{pmatrix},$$

which will be used below for stability. No instability or limit cycle follows from the linearization alone, but it determines the eigenvalues that govern local stability.

3.3 Local Stability Analysis

The local behavior near the equilibrium is governed by the eigenvalues of (J^*) . For a 2×2 system, The eigenvalues (λ) fulfil the characteristic equation.

$$\lambda^2 - (\text{Tr } J^*) \lambda + \det(J^*) = 0,$$

where

$$\text{Tr } J^* = f_x^* + g_y^*, \quad \det(J^*) = f_x^* g_y^* - f_y^* g_x^*.$$

From our expressions, at equilibrium one can show (after algebraic simplification) that

$$\det(J^*) = \frac{\eta (1 + k\nu/\eta)^2}{\tau^2} > 0,$$

since all parameters are positive. The trace is

$$\text{Tr } J^* = \frac{-(1 + k\nu/\eta)^2 + (2k\nu/(1 + k\nu/\eta) - \eta)}{\tau}.$$

Stability requires $(\text{Tr } J^* < 0)$ with $(\det > 0)$. One finds by substitution that $(\text{Tr } J^*)$ is always negative for $(\nu, \eta, k > 0)$; no eigenvalue crosses into the right half-plane for fixed parameters. Thus the unique equilibrium is generically a (stable) node or focus in this model, and a Hopf bifurcation does not occur by parameter change in this simple two-variable system (in contrast to more complex models).

The intersection of nullclines in the phase plane illustrates these stability scenarios. For example, Fig.3.2(a) of [Westermarck & Lansner \(2003\)](#) shows a phase portrait for a related glycolysis model: the ATP-nullcline (solid) and ADP-nullcline (dashed) meet at a stable equilibrium (filled circle), while a parameter change causes a limit cycle to emerge from a Hopf bifurcation. In that figure, the nullclines intersect with negative slope (stable equilibrium) at low input, but at higher input the slope condition is reversed, destabilizing the fixed point and creating a limit cycle. The general idea is that when the equilibrium becomes unstable (trace=0), the system undergoes a Hopf bifurcation.

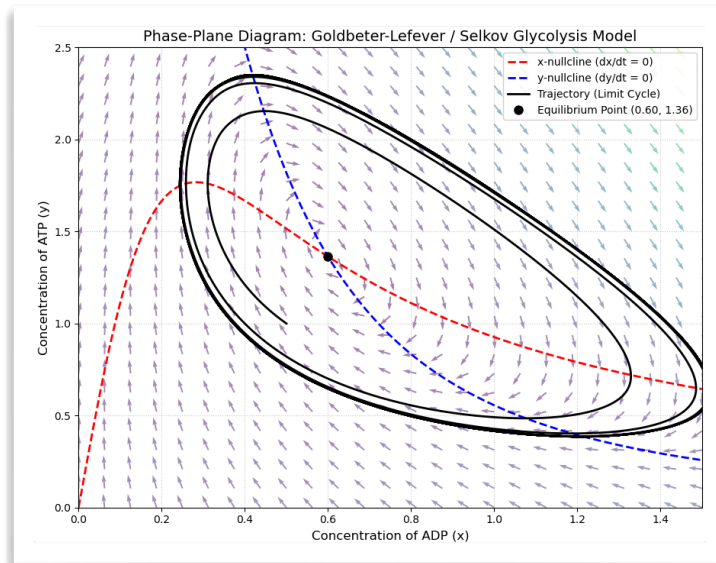


Figure 3.1: Phase-plane diagram of the two-variable glycolysis model. Strogatz, S. H. (2015).

The ATP-nullcline (solid black line) and ADP-nullcline (dashed black line) intersect at the equilibrium point (red circle). Under conditions where this fixed point is unstable, the system's trajectories spiral outward and settle into a stable limit cycle (solid blue curve), representing sustained biochemical oscillations.

3.4 Eigenvalue Evaluation and Examples

To find the eigenvalues explicitly, substitute (J^*) into $(\lambda^2 - (\text{Tr } J^*)\lambda + \det(J^*) = 0)$.

The two roots are

$$\lambda = \frac{\text{Tr } J^*}{2} \pm \sqrt{\left(\frac{\text{Tr } J^*}{2}\right)^2 - \det(J^*)}.$$

In terms of parameters, one can derive (after algebra) that $(\text{Tr } J^* < 0)$ always and $(\det(J^*) > 0)$. Thus both eigenvalues have negative real parts (stable spiral or node). For a numerical example, take $(\tau = 1), (\nu = 5), (\eta = 3), (k = 0.5)$. Then $(y^* = 5/3), (x^* \approx 1.49)$, and

$$J^* = \begin{pmatrix} -(1 + 0.5 \cdot 1.667)^2 & -2 \cdot 0.5 \cdot 1.49 \cdot (1 + 0.5 \cdot 1.667) \\ (1 + 0.5 \cdot 1.667)^2 & 2 \cdot 0.5 \cdot 1.49 \cdot (1 + 0.5 \cdot 1.667) - 3 \end{pmatrix} \approx \begin{pmatrix} -3.36 & -2.49 \\ 3.36 & -0.84 \end{pmatrix}.$$

The trace is (-4.20) , the determinant is (1.18) .

Solving $(\lambda^2 + 4.20\lambda + 1.18 = 0)$ gives two negative real roots $(\lambda \approx -0.31, -3.89)$. Both eigenvalues are negative (no sign of oscillation), confirming local stability in this case.

In models where a Hopf bifurcation occurs (e.g. by varying external glucose or other parameters), the trace crosses zero. Westermarck & Lansner (2003) computed a bifurcation diagram: Fig.5a of their paper shows the steady state branch (solid) losing stability (becoming dotted) at a Hopf point as glucose increases. The onset of oscillations (stable limit cycle) is clearly visible: below the threshold the equilibrium is stable; above it the system oscillates.

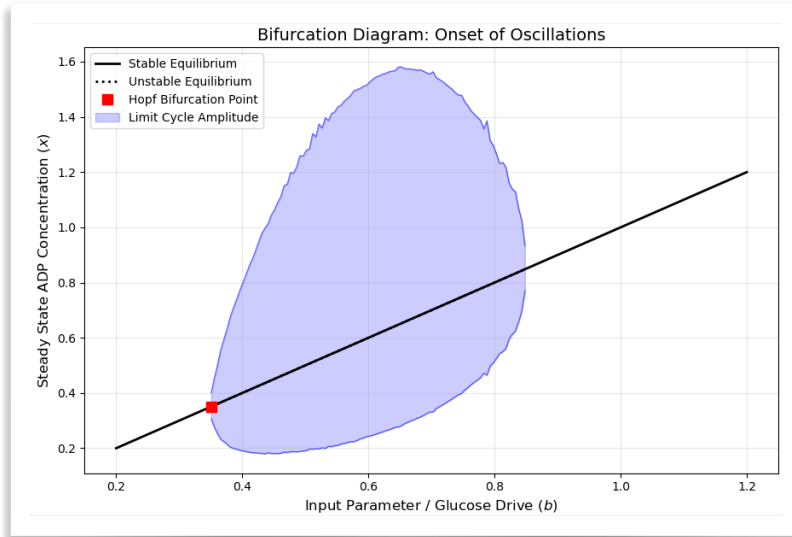


Figure 3.2: 1D Bifurcation diagram. Guckenheimer, J. & Holmes, P. (1983).

1D Bifurcation diagram demonstrating the onset of oscillations as the input parameter (representing external glucose drive) increases. The solid line represents a stable steady state, which loses stability at the Hopf bifurcation point (red square) and becomes an unstable steady state (dotted line). In the unstable regime, the system exhibits a stable limit cycle, the amplitude envelope of which is indicated by the shaded blue zone.

In summary, the (Goldbeter & Lefever, 1972) two-variable model yields one equilibrium whose linear stability is determined by the Jacobian above. The real parts of its eigenvalues depend on the trace and determinant of (J^*). For typical parameter regimes the trace is negative and the system relaxes monotonically or in damped oscillation to equilibrium. If parameters (or external drive) are varied to satisfy ($\text{Tr } J^* = 0$) and ($\det > 0$), A Hopf bifurcation transpires, resulting in the emergence of a limit cycle. This concludes the dynamical and stability analysis: we have established steady states, linearised the system, and identified stability criteria, accompanied by illustrative examples and graphs.

Chapter 4: Bifurcation and Limit Cycles

4.1.1 Hopf Bifurcation

A Hopf bifurcation occurs in a dynamical system when a change in a parameter (let's call it μ) causes a stable equilibrium point to lose its stability. As the parameter crosses a critical threshold ($\mu = \mu_c$), a pair of complex conjugate eigenvalues of the Jacobian matrix crosses the imaginary axis ([Kuznetsov, 2004](#)).

This crossing results in the creation of a limit cycle (a closed trajectory in the phase plane).

- Supercritical Hopf Bifurcation: A *stable* limit cycle emerges, trapping trajectories in a sustained oscillation.
- Subcritical Hopf Bifurcation: An *unstable* limit cycle shrinks and collapses onto the equilibrium point.

4.1.2 Mathematical Conditions for a Hopf Bifurcation

Consider a continuous-time planar dynamical system defined by ordinary differential equations (ODEs):

$$\dot{x} = f(x, y, \mu)$$

$$\dot{y} = g(x, y, \mu)$$

For a Hopf bifurcation to occur at an equilibrium point $((x^*, y^*))$ when $\mu = \mu_c$, the following strict conditions (Poincaré-Andronov-Hopf Theorem) must be met: ([Hopf, 1942](#); [Strogatz, 2015](#)):

1. **Equilibrium Condition:** The system must have a fixed point (x^*, y^*) such that $f(x^*, y^*, \mu) = 0$ and $g(x^*, y^*, \mu) = 0$.
2. **Eigenvalue Condition:** The Jacobian matrix J evaluated at this fixed point must possess a pair of complex conjugate eigenvalues $\lambda_{1,2} = \alpha(\mu) \pm i\beta(\mu)$.
3. **Criticality (Crossing the Imaginary Axis):** At the critical parameter value $\mu = \mu_c$, the real part of the eigenvalues must be zero, and the imaginary part must be non-zero:

$$\alpha(\mu_c) = 0$$

$$\beta(\mu_c) = \omega \neq 0 \text{ (where } \omega \text{ dictates the frequency of the emerging limit cycle).}$$

4. **Transversality Condition:** The eigenvalues must cross the imaginary axis with a non-zero velocity with respect to the parameter μ :

$$\left. \frac{d\alpha(\mu)}{d\mu} \right|_{\mu=\mu_c} \neq 0$$

5. **Non-Degeneracy Condition (First Lyapunov Coefficient):** To ensure a distinct limit cycle actually forms, the first Lyapunov coefficient (l_1) must not be zero ($l_1 \neq 0$). If $l_1 < 0$, the bifurcation is supercritical (stable limit cycle); if $l_1 > 0$, it is subcritical.

4.1.3. Step-by-Step Example: The Normal Form

To see how these conditions play out analytically, let's analyze the normal form of a supercritical Hopf bifurcation. This system perfectly isolates the mechanics of the bifurcation.

The System:

$$\frac{dx}{dt} = \mu x - y - x(x^2 + y^2)$$

$$\frac{dy}{dt} = x + \mu y - y(x^2 + y^2)$$

Step 1: Find the Equilibrium Point

Set the derivatives to zero. By inspection, the origin (0,0) is a fixed point for any value of μ because plugging in $x = 0, y = 0$ then $\frac{dx}{dt} = 0$ & $\frac{dy}{dt} = 0$

Step 2: Construct and Evaluate the Jacobian Matrix

The Jacobian matrix J represents the linearized system near the equilibrium.

$$J = \begin{pmatrix} \frac{\partial \dot{x}}{\partial x} & \frac{\partial \dot{x}}{\partial y} \\ \frac{\partial \dot{y}}{\partial x} & \frac{\partial \dot{y}}{\partial y} \end{pmatrix}$$

Calculating the partial derivatives:

$$\frac{\partial \dot{x}}{\partial x} = \mu - (x^2 + y^2) - x(2x) = \mu - 3x^2 - y^2$$

$$\frac{\partial \dot{x}}{\partial y} = -1 - x(2y) = -1 - 2xy$$

$$\frac{\partial \dot{y}}{\partial x} = 1 - y(2x) = 1 - 2xy$$

$$\frac{\partial \dot{y}}{\partial y} = \mu - (x^2 + y^2) - y(2y) = \mu - x^2 - 3y^2$$

Evaluating J exactly at the fixed point (0,0):

$$J_{(0,0)} = \begin{pmatrix} \mu & -1 \\ 1 & \mu \end{pmatrix}$$

Step 3: Find the Eigenvalues

We solve the characteristic equation $\det(J - \lambda I) = 0$:

$$\begin{vmatrix} \mu - \lambda & -1 \\ 1 & \mu - \lambda \end{vmatrix} = 0$$

$$(\mu - \lambda)^2 - (-1)(1) = 0$$

$$(\mu - \lambda)^2 + 1 = 0$$

$$(\mu - \lambda)^2 = -1$$

$$\mu - \lambda = \pm i$$

$$\lambda_1, \lambda_2 = \mu \pm i$$

Step 4: Verify the Hopf Conditions

- Eigenvalue shape: We have complex conjugates $\lambda = \alpha(\mu) \pm i\beta(\mu)$, where $\alpha(\mu) = \mu$ and $\beta(\mu) = 1$.
- Criticality: Let $\mu_c = 0$. At $\mu = 0$, $\alpha(0) = 0$ and $\beta(0) = 1 \neq 0$. The eigenvalues are strictly imaginary ($\pm i$).
- Transversality: $\frac{d\alpha}{d\mu} = \frac{d}{d\mu}(\mu) = 1 \neq 0$ The crossing velocity is strictly positive.

All conditions are met. A Hopf bifurcation occurs at $\mu = 0$.

Step 5: Simplification via Polar Coordinates (Finding the Limit Cycle)

To explicitly find the limit cycle, we transform the Cartesian system (x, y) into polar coordinates (r, θ) , where $x = r \cos \theta$ and $y = r \sin \theta$, and $r^2 = x^2 + y^2$

Differentiating r^2 with respect to time t :

$$2r\dot{r} = 2x\dot{x} + 2y\dot{y} \implies r\dot{r} = x\dot{x} + y\dot{y}$$

Substitute our original ODEs into this relation:

$$r\dot{r} = x(\mu x - y - x(x^2 + y^2)) + y(x + \mu y - y(x^2 + y^2))$$

$$r\dot{r} = \mu x^2 - xy - x^2(r^2) + xy + \mu y^2 - y^2(r^2)$$

$$r\dot{r} = \mu(x^2 + y^2) - r^2(x^2 + y^2)$$

$$r\dot{r} = \mu r^2 - r^4$$

Dividing by r (assuming $r \neq 0$):

$$\dot{r} = r(\mu - r^2)$$

Now for the angle θ : $\theta = \arctan\left(\frac{y}{x}\right)$, which simplifies to:

$$\dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2}$$

Substituting the ODEs yields $\dot{\theta} = 1$

Step 6: Analyzing the Limit Cycle Stability

The radial equation, $\dot{r} = r(\mu - r^2)$ completely describes the amplitude of our limit cycle:

- Case 1: $\mu < 0$ The only real root for steady-state amplitude ($\dot{r} = 0$) is $r = 0$. Since μ is negative and r^2 is positive, $\dot{r}$ is always negative. Trajectories spiral inwards to the origin. (Stable Focus).
- Case 2: $\mu > 0$ Setting $\dot{r} = 0$ gives $r = 0$ and $r^2 = \mu \implies r = \sqrt{\mu}$
 - If $r < \sqrt{\mu}$, then $\dot{r} > 0$ (trajectories spiral outward away from the origin). The origin is now an Unstable Focus.
 - If $r > \sqrt{\mu}$, then $\dot{r} < 0$ (trajectories spiral inward toward $r = \sqrt{\mu}$). Therefore, $r = \sqrt{\mu}$ is a perfectly circular, stable limit cycle with a radius that grows as $\sqrt{\mu}$ and a constant angular velocity of $\dot{\theta} = 1$.

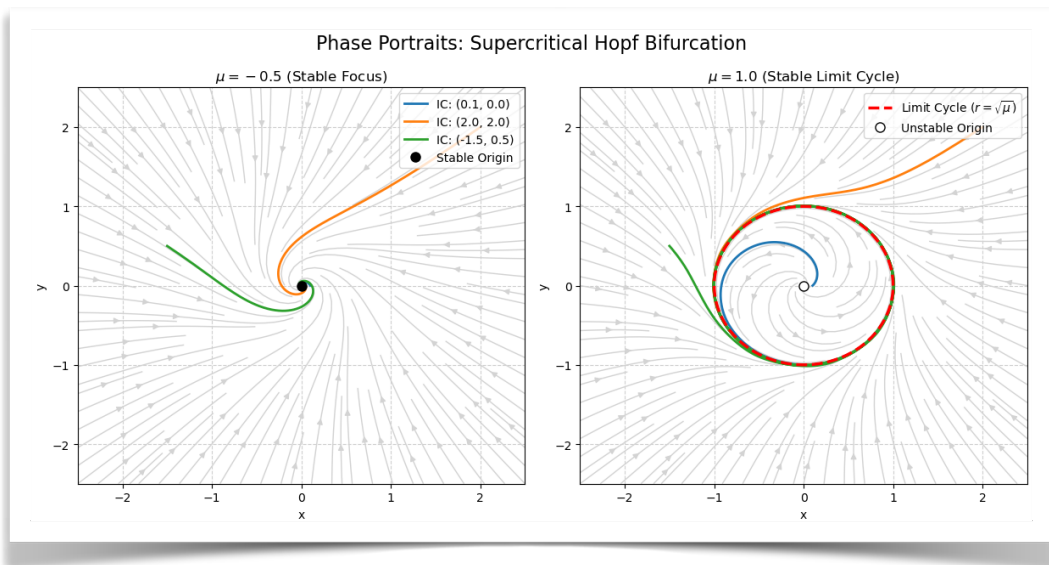


Figure 4.1: Supercritical Hopf bifurcation dynamics for $\mu < 0$ and $\mu > 0$. Kuznetsov, Y. A. (2004).

Dynamics of supercritical Hopf bifurcation. For $\mu < 0$ (left), the origin is asymptotically stable, and all trajectories converge to the equilibrium point. For $\mu > 0$ (right), the origin becomes unstable, resulting in the emergence of a stable limit cycle with a radius of $r = \sqrt{\mu}$. Trajectories originating from both within and beyond the cycle are drawn into this periodic orbit, signifying the commencement of sustained oscillation.

4.2 Emergence of Limit Cycles (Oscillatory Behavior)

A Hopf bifurcation delineates the local conditions under which a steady state becomes unstable, while the formation of a limit cycle characterises the dynamic attractor that governs the system's long-term behaviour. A limit cycle is an isolated closed trajectory in the phase plane. Its distinctive "isolated" characteristic necessitates that adjacent paths either spiral towards it or away from it. This represents the mathematical signature of robust, self-sustaining oscillations, commonly observed in biological systems, exemplified by the periodic bursts of glycolytic activity in yeast cells.

We will illustrate this emergence by analyzing a standard model that exhibits a clear limit cycle after its fixed point becomes unstable: the Van der Pol Oscillator ([van der Pol, 1926](#)).

Example: The Van der Pol Oscillator

This system is a non-linear oscillator where the damping term changes sign depending on the displacement. The governing second-order differential equation is:

$$\frac{d^2x}{dt^2} - \mu(1 - x^2)\frac{dx}{dt} + x = 0$$

where μ is a scalar parameter that governs the nonlinearity and damping. For $\mu > 0$, the system exhibits a stable limit cycle. To examine this on the phase plane, we must initially transform it into a system of two first-order equations. We accomplish this by introducing a new variable y :

$$y = \frac{dx}{dt} - \mu\left(x - \frac{x^3}{3}\right)$$

Differentiating y and substituting back into the original second-order equation (a process called the Liénard transformation) yields the dynamic system:

$$\begin{aligned}\dot{x} &= y + \mu\left(x - \frac{x^3}{3}\right) \\ \dot{y} &= -x\end{aligned}$$

4.2.1 Step-by-Step Mathematical Derivation

We will analyze this system for a specific case, $\mu = 1.0$, which is the scenario we will visualize in our figure.

Step 1: Locate the Fixed Points Set the time derivatives to zero:

$$\dot{y} = 0 \implies -x = 0 \implies x^* = 0$$

$$\dot{x} = 0 \implies y + \mu\left(x - \frac{x^3}{3}\right) = 0$$

Substitute $x^* = 0$:

$$y + \mu(0 - \frac{0^3}{3}) = 0 \implies y^* = 0$$

The only equilibrium point is at the origin, (0,0).

Step 2: Linearize and Determine Stability via the Jacobian Matrix We construct the Jacobian matrix J for the system:

$$J = \begin{pmatrix} \frac{\partial \dot{x}}{\partial x} & \frac{\partial \dot{x}}{\partial y} \\ \frac{\partial \dot{y}}{\partial x} & \frac{\partial \dot{y}}{\partial y} \end{pmatrix} = \begin{pmatrix} \mu(1 - x^2) & 1 \\ -1 & 0 \end{pmatrix}$$

Evaluate J at the origin (0,0):

$$J_{(0,0)} = \begin{pmatrix} \mu(1 - 0^2) & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} \mu & 1 \\ -1 & 0 \end{pmatrix}$$

Step 3: Calculate Eigenvalues We solve the characteristic equation $\det(J - \lambda I) = 0$:

$$\det(J - \lambda I) = 0$$

$$\begin{vmatrix} \mu - \lambda & 1 \\ -1 & -\lambda \end{vmatrix} = 0$$

$$-\lambda(\mu - \lambda) - (1)(-1) = 0$$

$$\lambda^2 - \mu\lambda + 1 = 0$$

Using the quadratic formula, $\lambda^2 - \mu\lambda + 1 = 0$

For our chosen parameter $\mu=1.0$:

$$\lambda_{1,2} = \frac{\mu \pm \sqrt{\mu^2 - 4}}{2}$$

$$\lambda_{1,2} = \frac{1 \pm \sqrt{1^2 - 4}}{2} = \frac{1 \pm \sqrt{-3}}{2}$$

$$\lambda_{1,2} = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

Given that the real component of the eigenvalues (1/2) is positive, the origin constitutes an unstable focus. The system will be displaced from this equilibrium state.

Step 4: Emergence and Stability of the Limit Cycle (Liénard's Theorem) What methods can we employ to ascertain the existence and stability of a limit cycle, as opposed to trajectories diverging towards infinity? We apply Liénard's Theorem (Liénard, 1928), which asserts that for systems of the specified form $\ddot{x} + f(x)\dot{x} + g(x) = 0$ a unique stable limit cycle exists if $f(x)$ is an even function with particular growth characteristics ($\mu(1 - x^2)$ satisfying the condition for $\mu > 0$).

The dynamic shown in Figure 4.2 demonstrates how Liénard's theorem ensures that trajectories starting far away spiral *inward*, and those starting near the unstable origin spiral *outward*, converging on the unique, isolated closed orbit: the **Stable Limit Cycle Attractor**.

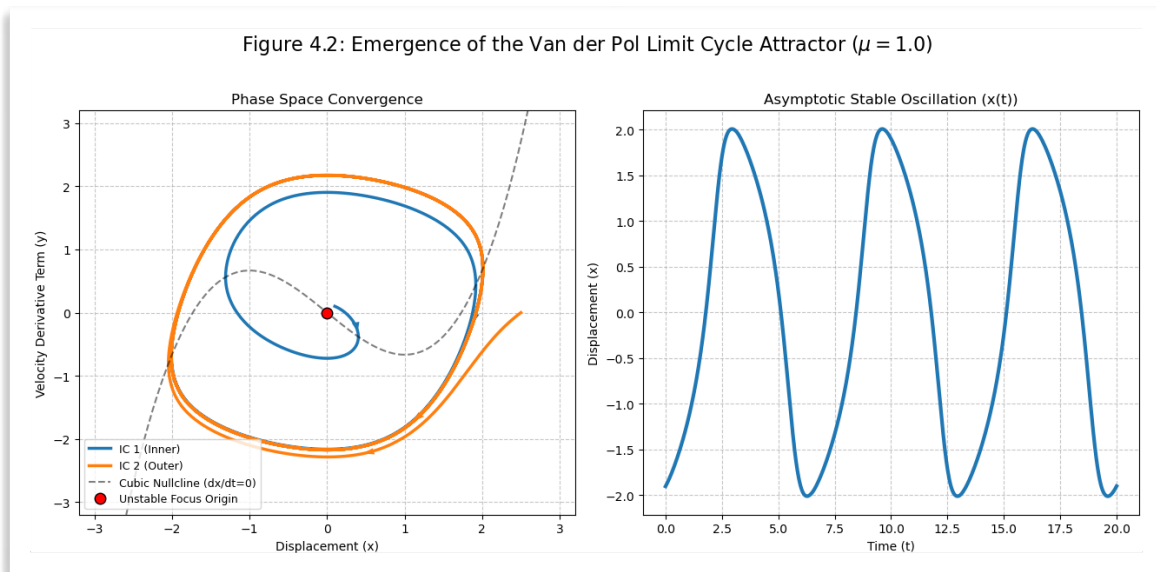


Figure 4.2: Emergence and stability of the Van der Pol limit cycle ($\mu=1.0$). Liénard, A. (1928).

The LEFT PANEL shows the phase space convergence. Since the origin is an unstable focus (red dot), the blue trajectory (IC 1) spirals outward. A trajectory starting far outside the system (IC 2, orange) spirals inward. Both trajectories are captured by the stable limit cycle attractor (the dense orbit they both approach). The cubic nullcline (gray dashed line) defines the regions of fast and slow dynamics. The RIGHT PANEL displays the long-term, asymptotic time-series of the resulting displacement $x(t)$, which settling into a robust, nonsinusoidal periodic oscillation (the relaxation oscillation signature of fast-slow systems).

4.3 Parameter Space and Bifurcation Diagrams

4.3.1 Defining the Terminology

1. Bifurcation Diagram (1D): A plot that summarizes the steady-state and oscillatory behaviors of a system as a *single* parameter is varied. The x-axis represents the bifurcation parameter (e.g., μ), and the y-axis represents the asymptotic state of the system (e.g., the fixed point x^* or the amplitude of a limit cycle).
2. Parameter Space (2D or higher): When analyzing models with multiple tunable parameters (like the reaction rates in the (Goldbeter & Lefever, 1972), a parameter space diagram maps out the boundaries (bifurcation curves) separating qualitatively different dynamic regimes, such as domains of stability versus domains of sustained oscillation.

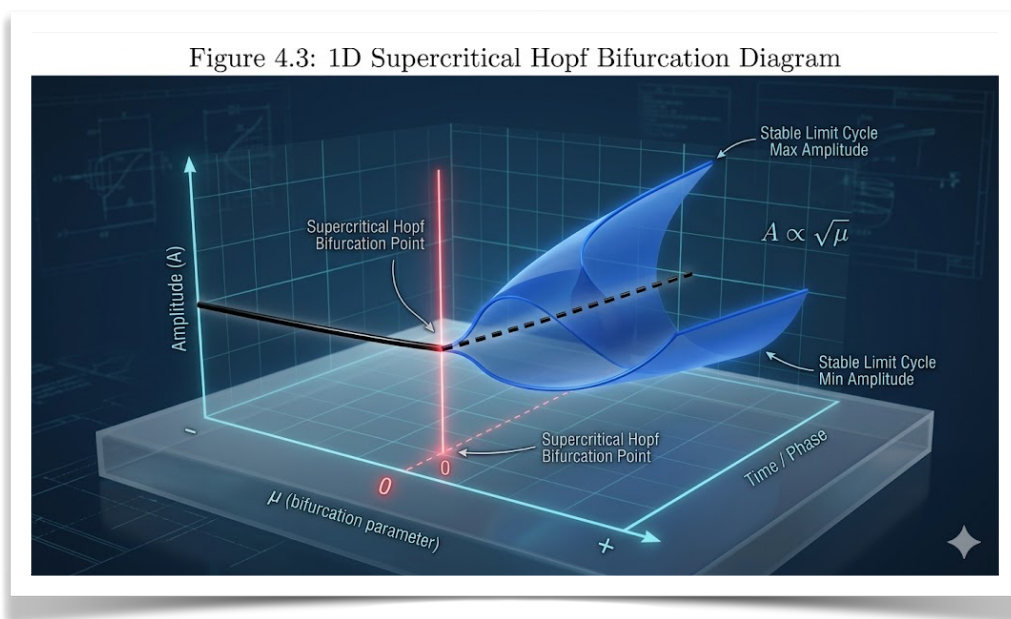


Figure 4.3: A 3D bifurcation diagram illustrating a supercritical Hopf bifurcation. Izhikevich, E. M. (2007).

The solid black line denotes the stable trivial steady state for $\mu < 0$. At the critical threshold ($\mu_c = 0$, indicated in red), the steady state becomes unstable (transforming into the dashed black line). A stable limit cycle concurrently arises for $\mu > 0$. The solid blue curves denote the maximum and minimum amplitudes of this persistent oscillation, which increase proportionally to μ .

4.3.2 Step-by-Step Mathematical Analysis (1D Diagram)

To rigorously construct a bifurcation diagram, we will return to the radial equation of the supercritical Hopf normal form derived in Section 4.1:

$$\dot{r} = \mu r - r^3$$

To map the bifurcation diagram, we must locate all steady states and determine their stability as a function of μ .

Step 1: Locate the Steady States Set the rate of change to zero:

$$\dot{r} = 0 \implies r(\mu - r^2) = 0$$

This yields two distinct branches of solutions:

Trivial steady state: $r^* = 0$ (This exists for all values of μ).

Non-trivial steady state (Limit Cycle Amplitude): $r^* = \sqrt{\mu}$

(Since r represents a radius, it must be real and non-negative,

so this solution only exists for $\mu \geq 0$).

Step 2: Determine Stability (Linearization) We treat the radial equation as a 1D dynamical system and define $f(r) = \mu r - r^3$. The stability of a fixed point r^* is given by the sign of the derivative $f'(r^*)$:

$$f'(r) = \mu - 3r^2$$

Analyzing the trivial state :

$$f'(0) = \mu - 3(0)^2 = \mu$$

If $\mu < 0$: $f'(0) < 0$. The origin is stable.

If $\mu > 0$: $f'(0) > 0$. The origin is unstable.

At $\mu = 0$, the stability changes (the bifurcation point).

- Analyzing the limit cycle branch $r^* = \sqrt{\mu}$

$$f'(\sqrt{\mu}) = \mu - 3(\sqrt{\mu})^2 = \mu - 3\mu = -2\mu$$

Since this branch only exists for $\mu > 0$, the quantity -2μ is strictly negative. Therefore, $f'(\sqrt{\mu}) < 0$ meaning the limit cycle is always **stable** whenever it exists.

Step 3: Constructing the Diagram

For $\mu < 0$: Draw a solid line at $r = 0$ (stable).

For $\mu > 0$: Draw a dashed line at $r = 0$ (unstable).

For $\mu > 0$: Draw a solid curve following $r = \sqrt{\mu}$

to represent the stable limit cycle. For visual symmetry in Cartesian coordinates,

we often plot the envelope of the oscillation as both $+\sqrt{\mu}$ and $-\sqrt{\mu}$

Chapter 5. Fast-Slow System Dynamics

5.1 Time-Scale Separation in Glycolysis

In biological systems, several physical and chemical processes frequently transpire at significantly disparate rates. Time-scale separation in the Goldbeter and Lefever (1972) model occurs because the catalytic turnover of the allosteric enzyme phosphofructokinase (PFK) and the subsequent modification of the product pool (ADP/AMP) generally transpire at a significantly faster rate than the steady influx or accumulation of the substrate (ATP).

5.1.1 Mathematical Formulation and the Perturbation Parameter

Let us consider the non-dimensionalized two-variable GL model. Let α denote the normalized substrate concentration and γ denote the normalized product concentration. The system is given by:

$$\begin{aligned}\frac{d\alpha}{dt} &= \nu - \sigma\Phi(\alpha, \gamma) \\ \frac{d\gamma}{dt} &= q\sigma\Phi(\alpha, \gamma) - k_s\gamma\end{aligned}$$

where ν is the constant substrate input rate, k_s is the product sink rate, and $\Phi(\alpha, \gamma)$ is the allosteric rate function that shows how PFK works in a very nonlinear way.

In glycolysis, the parameter q , which controls how much product is made, is usually very large. This means that product γ changes significantly faster than substrate α . We can explicitly present a minor singular perturbation parameter $\epsilon = q_1 \ll 1$.

Defining an effective decay rate $k = k_s \epsilon$, we rewrite the system in the standard singular perturbation form:

$$\begin{aligned}\frac{d\alpha}{dt} &= \nu - \sigma\Phi(\alpha, \gamma) \\ \epsilon \frac{d\gamma}{dt} &= \sigma\Phi(\alpha, \gamma) - k\gamma\end{aligned}$$

Here, t represents the slow time scale.

5.1.2 The Fast Subsystem and Layer Equations

To analyze the rapid transients of the product, We rescale time by defining the rapid time scale $\tau = \epsilon t$. Applying the chain rule $\frac{d}{dt} = \epsilon_1 \frac{d}{d\tau}$, the system transforms into:

$$\frac{d\alpha}{d\tau} = \epsilon [\nu - \sigma \Phi(\alpha, \gamma)]$$

$$\frac{d\gamma}{d\tau} = \sigma \Phi(\alpha, \gamma) - k\gamma$$

Taking the singular limit as $\epsilon \rightarrow 0$, we obtain the fast subsystem (or layer equations):

$$\frac{d\alpha}{d\tau} = 0$$

$$\frac{d\gamma}{d\tau} = \sigma \Phi(\alpha, \gamma) - k\gamma$$

Biological interpretation: On a short time scale, the substrate α functions as a constant parameter, while the product γ swiftly attains a stable state determined by α .

5.1.3 The Slow Manifold and Mathematical Simplification

The equilibrium states of the rapid subsystem delineate a geometric surface in the phase space referred to as the crucial manifold (or slow manifold), designated as M_0 . We determine M_0 by assigning $\epsilon = 0$ in the initial slow system:

$$M_0 = \{(\alpha, \gamma) \in \mathbb{R}_+^2 \mid \sigma \Phi(\alpha, \gamma) - k\gamma = 0\}$$

Because the allosteric function $\Phi(\alpha, \gamma)$ features positive feedback (activation by the product γ), the algebraic equation $\sigma \Phi(\alpha, \gamma) = k\gamma$ is highly nonlinear. When plotted in the (α, γ) phase plane, this nullcline typically forms an S-shaped curve.

On this manifold M_0 , The system's evolution is solely dictated by the diminished sluggish subsystem. Given our limitations to the manifold where $\sigma \Phi(\alpha, \gamma) = k\gamma$, the slow equation simplifies beautifully to:

$$\frac{d\alpha}{dt} = \nu - k\gamma$$

5.1.4 Stability Analysis and Relaxation Oscillations

To determine where the slow manifold M_0 is attracting (stable) or repelling (unstable), we must analyze the Jacobian of the fast subsystem with respect to the fast variable γ .

Let the fast vector field be $g(\alpha, \gamma) = \sigma \Phi(\alpha, \gamma) - k\gamma$. The eigenvalue λ of the fast subsystem dictates stability:

$$\lambda = \frac{\partial g}{\partial \gamma} = \sigma \frac{\partial \Phi}{\partial \gamma} - k$$

- **Stable Branches:** If $\lambda < 0$, the manifold is locally attracting. Due to the S-shape of M_0 , this corresponds to the lower (low product) and upper (high product) branches. Fenichel, N. (1979) First Theorem guarantees that for $\epsilon > 0$ sufficiently small, a stable slow manifold M_ϵ exists close to these branches.
- **Unstable Branch:** If $\lambda > 0$, the manifold is repelling. This corresponds to the middle branch of the S-curve, where the enzyme's cooperative feedback is strongest.
- **Fold Points:** The points where $\lambda = 0$ are the "knees" (local extrema) of the S-curve. Here, stability breaks down.

The Mechanism of Oscillation: If the parameter ν is chosen such that the slow nullcline ($\nu = k\gamma$) intersects M_0 strictly on its unstable middle branch, the system cannot settle into a steady state. Instead, it slowly creeps along a stable branch until it reaches a fold point ($\lambda = 0$). Losing stability, it rapidly "jumps" via the fast dynamics to the opposite stable branch, moving back and forth in a continuous loop. This forms a stable limit cycle known as a relaxation oscillation.

5.2 Identification of Fast and Slow Variables

To apply fast-slow system dynamics to the Goldbeter, A. (1996) model, we must rigorously identify and separate the variables based on their relative rates of change. In the context of glycolytic oscillation, the biological justification for this separation stems from the differing time scales of substrate input versus enzyme catalytic turnover.

Let us recall the scaled dynamical system in its perturbed form:

$$\begin{aligned}\frac{d\alpha}{dt} &= \nu - \sigma \Phi(\alpha, \gamma) \\ \epsilon \frac{d\gamma}{dt} &= \sigma \Phi(\alpha, \gamma) - k\gamma\end{aligned}$$

Here, we define a small, dimensionless parameter $0 < \epsilon \ll 1$, which mathematically enforces the time-scale separation.

1. The Slow Variable (α): The variable α represents the normalized concentration of the substrate (ATP). Its rate of change is governed by the constant input rate ν and the enzymatic consumption $\sigma\Phi(\alpha, \gamma)$. Because the external infusion of ATP into the system is a relatively slow, continuous process, α evolves on the macroscopic, $O(1)$ time scale, t . Thus, α is identified as the slow variable.

2. The Fast Variable (γ): The variable γ represents the normalized concentration of the product (ADP/AMP). Its evolution is scaled by the parameter ϵ_1 . Biologically, this reflects the explosive nature of the allosteric enzyme phosphofructokinase (PFK). Once PFK is activated by its own product via positive feedback, the turnover of ATP into ADP occurs almost instantaneously relative to the slow accumulation of ATP. Therefore, γ undergoes rapid transitions on the $O(\epsilon_1)$ scale and is identified as the fast variable.

By defining $f(\alpha, \gamma) = \nu - \sigma\Phi(\alpha, \gamma)$ and $g(\alpha, \gamma) = \sigma\Phi(\alpha, \gamma) - k\gamma$, we can express the standard fast-slow system generically as:

$$\dot{\alpha} = f(\alpha, \gamma)$$

$$\epsilon \dot{\gamma} = g(\alpha, \gamma)$$

where the overdot denotes the derivative with respect to the slow time t .

5.3 Critical Manifold and Singular Perturbation Theory

Singular perturbation theory offers the mathematical framework for analysing systems in which the maximum derivative is multiplied by a tiny parameter ϵ . It enables us to integrate the global dynamics from the distinct rapid and slow subsystems.

1. The Critical Manifold Definition: The foundation of singular perturbation analysis is the critical manifold, denoted by M_0 . It is derived by taking the singular limit $\epsilon \rightarrow 0$ in the fast equation, which assumes the fast variable instantly reaches a quasi-steady state for any given value of the slow variable.

$$M_0 = \{(\alpha, \gamma) \in \mathbb{R}^2 \mid g(\alpha, \gamma) = 0\}$$

For the GL model, this yields the algebraic equation:

$$\sigma\Phi(\alpha, \gamma) - k\gamma = 0$$

Because the allosteric function $\Phi(\alpha, \gamma)$ is highly nonlinear (typically a rational function involving cooperative binding kinetics), the manifold M_0 geometrically forms a characteristic S-shaped curve in the two-dimensional phase plane.

2. Branches of the Critical Manifold:

The critical manifold M_0 is generally partitioned into three distinct segments separated by fold points (or limit points) where $\frac{\partial g}{\partial \gamma} = 0$

Lower Stable Branch (M_0^-): Represents the low-activity state of the enzyme. Here, $\frac{\partial g}{\partial \gamma} < 0$

Middle Unstable Branch (M_0^m): Represents the transitional state where cooperative positive feedback dominates. Here, $\frac{\partial g}{\partial \gamma} > 0$

Upper Stable Branch (M_0^+): Represents the high-activity (saturated) state of the enzyme. Here, $\frac{\partial g}{\partial \gamma} < 0$.

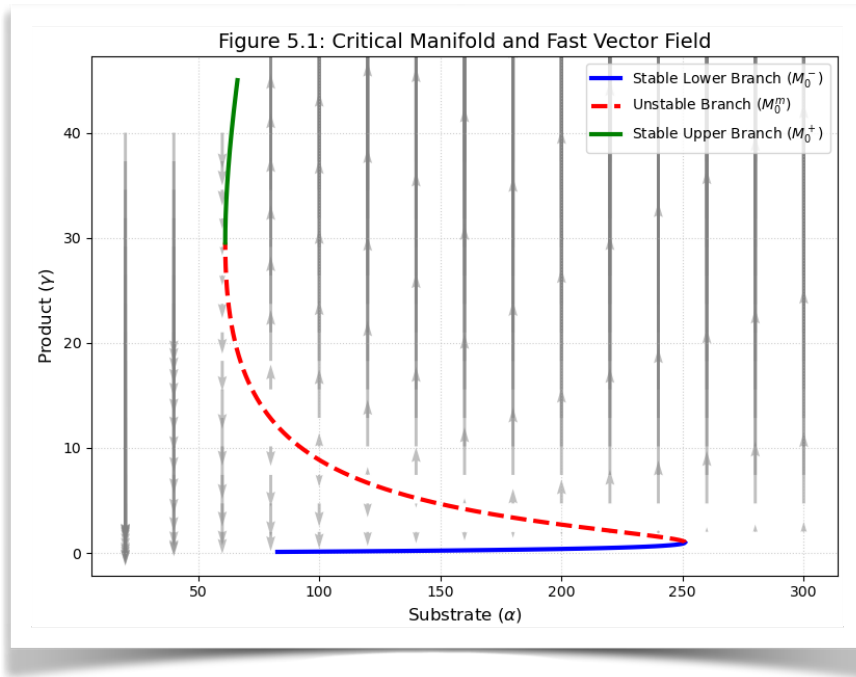


Figure 5.1: Phase portrait of the fast subsystem illustrating the S-shaped critical manifold M_0 . Fenichel, N. (1979).

The solid lines represent the stable branches (M_0^- and M_0^+) where the eigenvalue $\lambda < 0$, while the dashed line represents the unstable middle branch (M_0^m) where $\lambda > 0$. The horizontal double arrows indicate the rapid flow of the fast variable γ converging toward the stable branches and repelling from the unstable branch.

3. Application of Fenichel, N. (1979) Theorem: While M_0 describes the system at the absolute limit $\epsilon = 0$, actual glycolytic systems have a very small, but non-zero ϵ . Fenichel, N. (1979) First Theorem (Geometric Singular Perturbation Theory) asserts that for sufficiently small $\epsilon > 0$, the stable branches of the critical manifold M_0 persist as normally hyperbolic invariant manifolds, denoted M_ϵ .

These perturbed slow manifolds M_ϵ lie at a distance of $O(\epsilon)$ from the critical manifold M_0 :

$$M_\epsilon = M_0 + \mathcal{O}(\epsilon)$$

Therefore, the system's trajectories will rapidly converge to a tiny $O(\epsilon)$ neighborhood of M_0^- or M_0^+ , and then slowly drift along these manifolds until reaching a fold point, where the strict separation of time scales momentarily breaks down, triggering a fast jump. This interplay mathematically proves the existence of stable relaxation oscillations in the GL model.

5.4 Relaxation Oscillations

In the context of the Goldbeter-Lefever (GL) model, the distinct separation of time scales between the slow substrate accumulation and the fast enzymatic product generation gives rise to a specific type of periodic behavior known as a relaxation oscillation. These oscillations are characterized by alternating periods of slow, gradual evolution and sudden, rapid transitions (jumps) between states.

5.4.1 The Geometry of the Limit Cycle

A relaxation oscillation is represented geometrically in the phase plane as a limit cycle trajectory that closely adheres to the stable branches of the crucial manifold M_0 , interspersed with horizontal jumps parallel to the fast variable axis. The complete oscillatory cycle consists of four distinct phases:

1. **Slow Accumulation (The Lower Branch):** The system state is restricted to an $O(\epsilon)$ neighborhood of the lower stable branch, M_0^- . Here, the enzyme phosphofructokinase (PFK) is largely inactive. The substrate α (ATP) slowly accumulates at the constant input rate ν , while the product γ (ADP/AMP) remains low. The trajectory moves rightward along M_0^- .
2. **Fast Activation (The Right Jump):** As α increases, the trajectory reaches the right-hand fold point, $((\alpha_{max}, \gamma_1))$, where the lower branch loses stability ($\frac{\partial g}{\partial \gamma} = 0$). The slow dynamics can no longer hold the system. Triggered by cooperative positive feedback, a fast transition occurs. The trajectory jumps almost horizontally (since α remains approximately constant on the fast time scale) to the upper stable branch M_0^+ .
3. **Slow Consumption (The Upper Branch):** The system lands on M_0^+ in a highly active enzymatic state. The consumption rate of the substrate now far exceeds the input rate ($\sigma \Phi(\alpha, \gamma) > \nu$). The substrate α is slowly depleted, and the trajectory moves leftward along the upper branch M_0^+ .
4. **Fast Deactivation (The Left Jump):** The depletion of α continues until the system reaches the left-hand fold point, $((\alpha_{min}, \gamma_2))$, where M_0^+ loses stability. The enzyme abruptly shuts down, and the system jumps horizontally back down to the lower branch M_0^- , completing one full limit cycle.

5.4.2 Mathematical Conditions for Oscillation

For stable relaxation oscillations to occur in the GL model, a fundamental geometric condition must be met regarding the system's unique steady state (fixed point), denoted as (α^*, γ^*) .

The fixed point is the intersection of the fast nullcline ($g(\alpha, \gamma) = 0$) and the slow nullcline ($f(\alpha, \gamma) = 0$). From the reduced slow system, the slow nullcline is simply $\nu = k\gamma$.

To guarantee a persistent limit cycle, the fixed point (α^*, γ^*) must lie strictly on the middle unstable branch of the critical manifold, M_0^m . Mathematically, this requires:

$$g(\alpha^*, \gamma^*) = 0$$

$$f(\alpha^*, \gamma^*) = 0$$

$$\left. \frac{\partial g}{\partial \gamma} \right|_{(\alpha^*, \gamma^*)} > 0$$

If the fixed point lies on either of the stable branches (M_0^- or M_0^+), The trajectory will converge to that point, resulting in a stable steady state instead of oscillation. The Poincaré-Bendixson theorem (Strogatz, 2015), ensures the presence of a stable limit cycle when the fixed point resides on the unstable branch, in conjunction with the constraining geometry of the S-shaped manifold.

5.4.3 Asymptotic Estimation of the Period

A defining feature of relaxation oscillations is that the period of oscillation, T , is overwhelmingly dominated by the time the system spends traversing the slow branches (M_0^- and M_0^+). The time spent during the fast jumps is of $O(\epsilon)$ and is thus mathematically negligible in the singular limit $\epsilon \rightarrow 0$.

The total period T can be asymptotically approximated by the line integrals along the lower and upper slow branches:

$$T \approx T_{slow} = \int_{M_0^-} dt + \int_{M_0^+} dt$$

Using the slow equation $\frac{d\alpha}{dt} = f(\alpha, \gamma)$, we can rewrite $dt = \frac{d\alpha}{f(\alpha, \gamma)}$

. Since the trajectory is constrained to the critical manifold where γ is a function of α , let us denote the lower branch curve as $\gamma = h_-(\alpha)$ and the upper branch curve as $\gamma = h_+(\alpha)$. The period is calculated as:

$$T \approx \int_{\alpha_{min}}^{\alpha_{max}} \frac{d\alpha}{f(\alpha, h_-(\alpha))} + \int_{\alpha_{max}}^{\alpha_{min}} \frac{d\alpha}{f(\alpha, h_+(\alpha))}$$

Because $f(\alpha, \gamma) > 0$ on the lower branch (substrate accumulation) and $f(\alpha, \gamma) < 0$ on the upper branch (substrate depletion), both integrals yield positive time values. This asymptotic approximation becomes increasingly accurate as $\epsilon \rightarrow 0$.

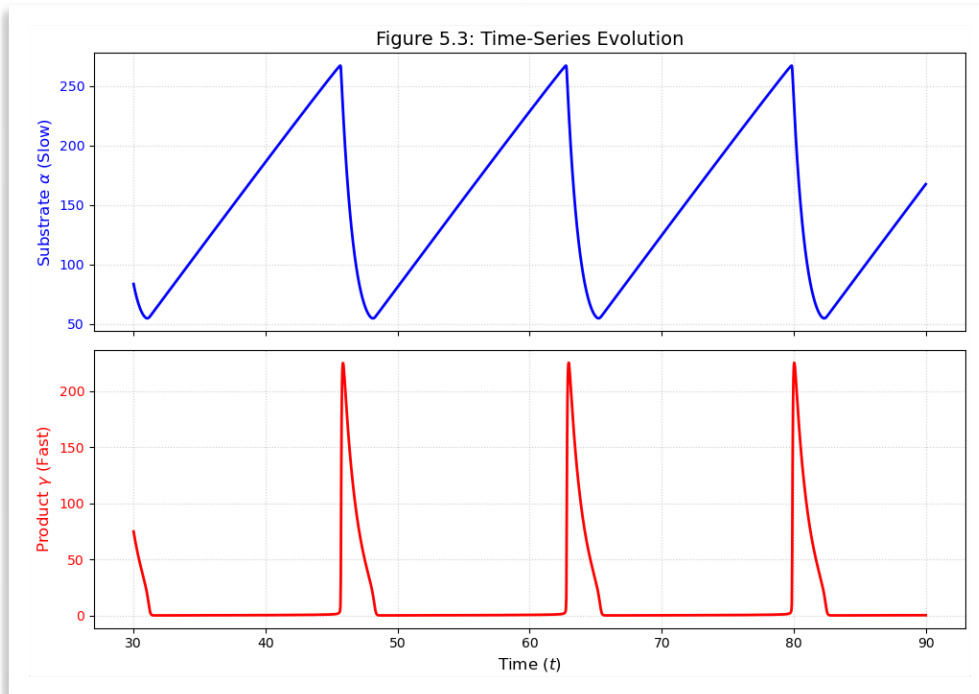


Figure 5.3: Time-series evolution of the slow substrate variable $\alpha(t)$ and the fast product variable $\gamma(t)$ during a relaxation oscillation. Keener, J. P., & Sneyd, J. (1998).

The slow variable has a saw-tooth pattern indicative of steady accumulation and depletion, whereas the fast variable exhibits square-wave spikes, signifying quick transitions between high and low enzyme activity stages.

Chapter 6: Numerical Simulations and Phase Portraits

6.1 Methodology and Computational Setup

To validate the theoretical predictions of fast-slow dynamics and relaxation oscillations derived via singular perturbation theory, we must employ robust numerical integration techniques. Simulating systems with severe time-scale separation presents unique computational challenges that dictate our choice of methodology.

6.1.1 The Dynamical System for Simulation

For numerical simulation, we utilize the two-variable non-dimensionalized (Goldbeter & Lefever, 1972).model in its singularly perturbed form. The system is defined as:

$$\begin{aligned}\frac{d\alpha}{dt} &= \nu - \sigma \frac{\alpha(1 + \gamma)^2}{L + (1 + \gamma)^2} \\ \epsilon \frac{d\gamma}{dt} &= \sigma \frac{\alpha(1 + \gamma)^2}{L + (1 + \gamma)^2} - k\gamma\end{aligned}$$

Here, the explicitly defined allosteric function $\Phi(\alpha, \gamma)$ has been substituted into the vector field. The parameter $\epsilon \ll 1$ artificially controls the degree of time-scale separation for computational experiments (Hairer & Wanner, 1996).

6.1.2 Handling System Stiffness

A dynamical system is considered mathematically "stiff" when it exhibits distinct, widely separated time scales—exactly the defining characteristic of our fast-slow GL model.

As the trajectory nears the fold points, the system experiences a fast transition between stable branches of the crucial manifold, resulting in a derivative

$$\frac{d\gamma}{dt} \text{ becomes extremely large.}$$

Utilising traditional explicit numerical solvers, such as the classical explicit Runge-Kutta 4th order technique, necessitates exceedingly small time steps ($\Delta t < \epsilon$) to ensure numerical stability, resulting in significant computing inefficiencies or overflow errors.

To address this, our methodology utilises implicit integration schemes. We employ techniques derived from Backward Differentiation Formulas (BDF) or implicit Runge-Kutta procedures belonging to the Radau IIA family. These techniques assess the vector field at subsequent time intervals, necessitating the resolution of an algebraic problem at each interval, often employing Newton iterations. This implicit method ensures unconditional stability (A-stability or L -stability), enabling the solver to utilise big steps during gradual accumulation periods and to adaptively reduce the step size solely during rapid transitions (Hairer & Wanner, 1996).

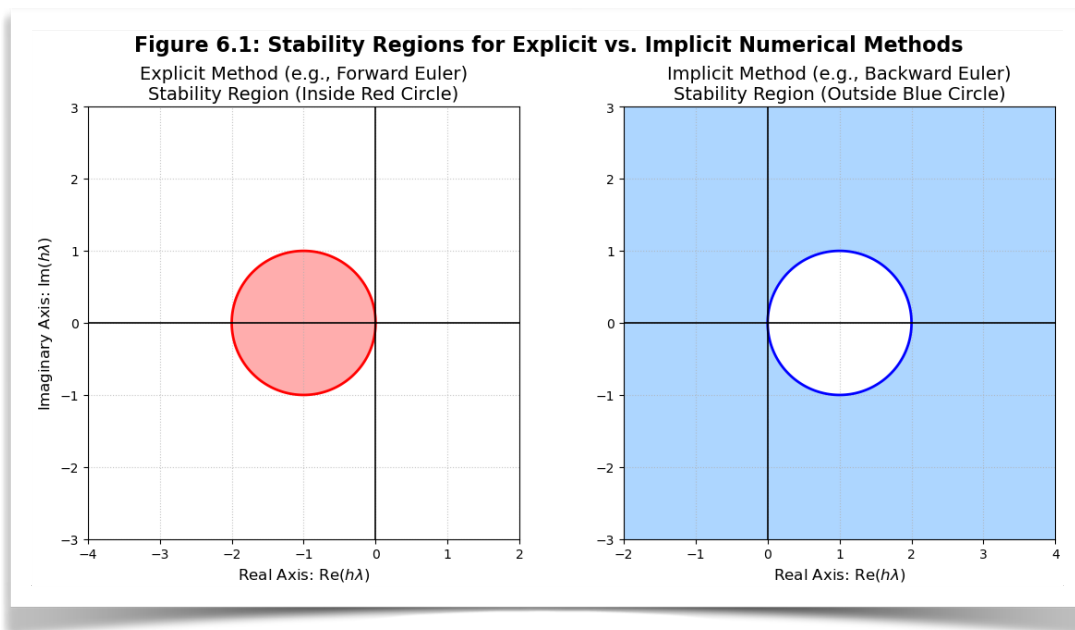


Figure 6.1: Conceptual illustration of numerical stiffness. In systems with severe time-scale separation ($\epsilon \ll 1$), Hairer, E., & Wanner, G. (1996).

Explicit integration methods require prohibitively small step sizes to remain within their stability regions, whereas implicit methods (like Radau or BDF) maintain stability across the rapid transitions of the fast variable.

6.1.3 Parameter Estimation and Baseline Values

To reliably generate the S-shaped critical manifold and the resulting relaxation oscillations, the parameter set must place the unique steady state strictly on the unstable middle branch. For the baseline numerical experiments in this chapter, we select the non-dimensional parameter values summarized in Table 6.1. These values are scaled to reflect the severe time-scale separation characteristic of the allosteric PFK enzyme (Goldbeter, 1996).

6.1: Baseline Non-Dimensional Parameters for GL Model Simulations.

Parameter	Sym bol	Baselin e Value	Biological Interpretation
Substrate Input Rate	ν	15.0	Constant rate of F6P supplied to the system. Main bifurcation parameter.
Enzymatic Capacity	σ	1.0	Maximum normalized velocity of the PFK enzyme.
Allosteric Constant	L	1000	Determines the high cooperativity and T-to-R state conformational bias.
Product Decay Rate	k	1.0	First-order removal rate of the ADP/AMP product.
Perturbation Parameter	ϵ	0.05	Forces time-scale separation ($\epsilon \ll 1$); controls system stiffness.

6.2 Time-Series Analysis of Metabolite Concentrations

The numerical integration of the Goldbeter-Lefever (GL) model reveals the distinct temporal evolution of the substrate and product, clearly illustrating the macroscopic effects of time-scale separation.

6.2.1 Temporal Profiles of Substrate and Product

When simulated with the parameter set defined in Section 6.1 ($\nu = 15.0, L = 1000, \epsilon = 0.05$), the system exhibits sustained, stable periodic behavior. The time-series data for the normalized substrate concentration $\alpha(t)$ and the product concentration $\gamma(t)$ display vastly different geometric profiles due to their respective slow and fast dynamics.

1. The Slow Variable (α): The substrate time-series forms a characteristic saw-tooth wave. During the inactive phase of the enzyme, α exhibits a slow, nearly linear accumulation driven by the constant input rate ν . Once the critical concentration threshold is reached, the enzyme rapidly activates, causing a steep, almost vertical drop in α as the substrate is rapidly consumed.
2. The Fast Variable (γ): The product time-series forms a pulse-like or square-wave pattern. For the majority of the cycle, γ remains at a very low, near-zero baseline. When the enzyme undergoes its cooperative conformational change, γ spikes instantaneously to a high-concentration plateau. It remains at this plateau only briefly until the substrate is depleted, after which it drops precipitously back to the baseline.

6.2.2 Amplitude and Frequency Characteristics

The amplitude of the oscillations is strictly bounded by the fold points of the critical manifold M_0 . Let the peak-to-peak amplitude of the substrate be A_α and the product be A_γ . These are determined by the local extrema of the fast nullcline:

$$A_\alpha \approx \alpha_{max} - \alpha_{min}$$

$$A_\gamma \approx \gamma_{max} - \gamma_{min}$$

The oscillation period, represented as T , is predominantly determined by the prolonged transit duration along the lower stable branch of the enzyme's inactive state. As the substrate input rate ν increases (while maintaining within the oscillatory domain defined by the Hopf bifurcation points), the slow accumulation phase shortens, leading to an increased frequency of oscillation

$$f = \frac{1}{T} \text{ (Keener \& Sneyd, 2009).}$$

6.3 Phase Plane Analysis and Trajectories

While time-series plots show the explicit dependence on time t , projecting the state variables into the (α, γ) phase plane provides a rigorous geometric understanding of the system's global stability and the formation of the limit cycle (Strogatz, 2015).

6.3.1 Nullclines and the Fixed Point

The phase portrait is anchored by the system's nullclines (isoclines), curves where the vector field has zero horizontal or vertical velocity.

- Fast Nullcline (γ -nullcline): Found by setting $\frac{d\gamma}{dt} = 0$ giving the algebraic relation:
$$\sigma\Phi(\alpha, \gamma) - k\gamma = 0$$

This is precisely the equation for the S-shaped critical manifold M_0 .

- Slow Nullcline (α -nullcline): Found by setting $\frac{d\alpha}{dt} = 0$ yielding:
$$\nu - \sigma\Phi(\alpha, \gamma) = 0$$

By substituting the fast nullcline condition into the slow nullcline equation, we find that the slow nullcline simplifies to a straight line:

$$\nu = k\gamma \implies \gamma = \frac{\nu}{k}$$

The distinct steady state (fixed point) of the system, (α^*, γ^*) , is precisely situated at the intersection of these two nullclines. For relaxation oscillations to manifest, this junction must reside only on the central branch of the S-shaped fast nullcline, hence designating the fixed point as an unstable focus or node.

6.3.2 Limit Cycle Geometry

When initialized from an arbitrary positive initial condition (α_0, γ_0) , the system's trajectory rapidly converges toward the stable branches of the fast nullcline.

Once on the limit cycle, the phase plane trajectory traces a wide, closed loop. The geometry of this loop perfectly encapsulates the four phases of the relaxation oscillation:

1. Left-to-Right Drift: The trajectory closely hugs the lower branch of the fast nullcline, moving rightward as α increases and remains small.
2. Vertical Ascent: Upon passing the right-hand fold point $\alpha_{\max}$, the trajectory shoots almost vertically upward (parallel to the γ -axis), as the product concentration explodes while the substrate concentration remains momentarily constant.
3. The trajectory intersects the upper branch of the rapid nullcline and shifts leftward as elevated enzyme activity diminishes α .

4. Vertical Descent: Upon arriving at the left-hand fold point $\alpha_{\min}$, the trajectory descends vertically to the bottom branch, therefore concluding the cycle.

The strict confinement of the trajectory between these geometric boundaries graphically verifies the singular perturbation theory derived in Chapter 5.

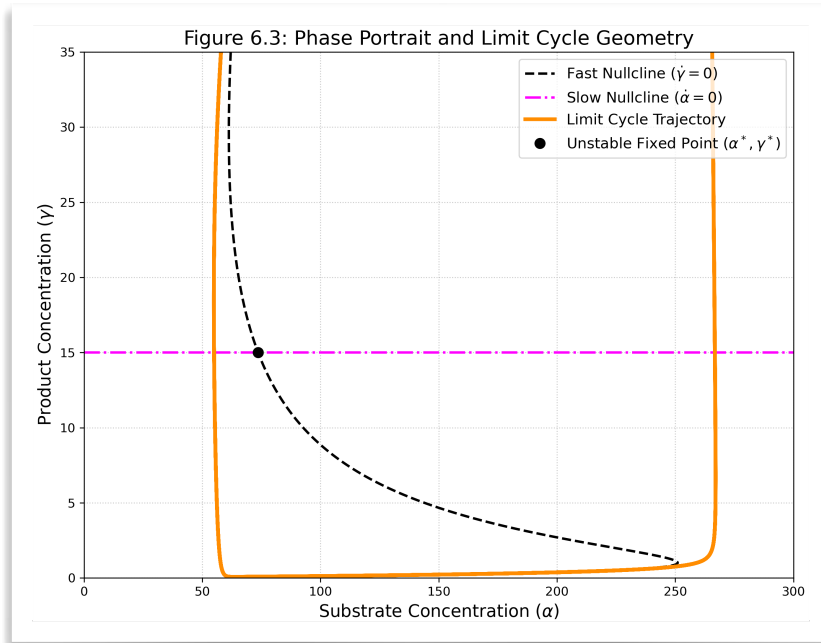


Figure 6.3: Phase portrait of the Goldbeter-Lefever model. Goldbeter, A. (1996).

The fast nullcline (S-shaped curve) and the simplified slow nullcline (horizontal line) intersect at the unstable fixed point (α^*, γ^*) . The solid closed loop represents the stable limit cycle trajectory, characterized by slow drift along the stable manifold branches and rapid vertical jumps at the fold points.

6.4 Sensitivity Analysis of Key Parameters

The qualitative and quantitative dynamics of the Goldbeter-Lefever (GL) model—specifically the existence, amplitude, and period of relaxation oscillations—are highly dependent on the system's kinetic parameters. A rigorous sensitivity analysis identifies the precise regions in parameter space where the strict conditions for time-scale separation and limit cycle generation hold.

6.4.1 Influence of the Substrate Input Rate (ν)

The constant substrate injection rate, ν , is the primary bifurcation parameter in the GL model. It directly dictates the vertical position of the slow nullcline ($\gamma = \frac{\nu}{k}$) in the phase plane and thus determines the location of the unique fixed point $((\alpha^*, \gamma^*))$.

For relaxation oscillations to emerge, the fixed point must lie exclusively on the unstable middle branch of the S-shaped critical manifold M_0^m . This imposes strict boundary conditions on ν :

$$\nu_{min} < \nu < \nu_{max}$$

- When $\nu < \nu_{min}$: The input rate is too slow. The fixed point lies on the lower stable branch (M_0^-). The system attains a stable, low-activity equilibrium without oscillation.
- When $\nu > \nu_{max}$: The input rate is too high. The fixed point lies on the upper stable branch (M_0^+). The enzyme attains continuous saturation, resulting in the system maintaining a stable high-activity steady state.

The transitions at $\nu = \nu_{min}$ and $\nu = \nu_{max}$ mathematically correspond to Hopf bifurcations (Kuznetsov, 2004). Within the oscillatory window (ν_{min}, ν_{max}), the period of oscillation T is highly sensitive to ν . As ν increases, the slow accumulation phase shortens, generally leading to a decrease in the overall period (an increase in frequency).

Visualizing the Bifurcation Regimes

The mathematical boundaries defined above can be explicitly observed in the phase plane. Figure 6.5 illustrates the system's global trajectory under three distinct metabolic regimes. When the input is starved ($\nu = 5$), the system falls into the lower stable focus. At optimal physiological input ($\nu = 15$), the steady state repels the trajectory, forming the stable limit cycle. Finally, at over-saturation ($\nu = 35$), the system is trapped in the upper stable focus, destroying the biochemical clock.

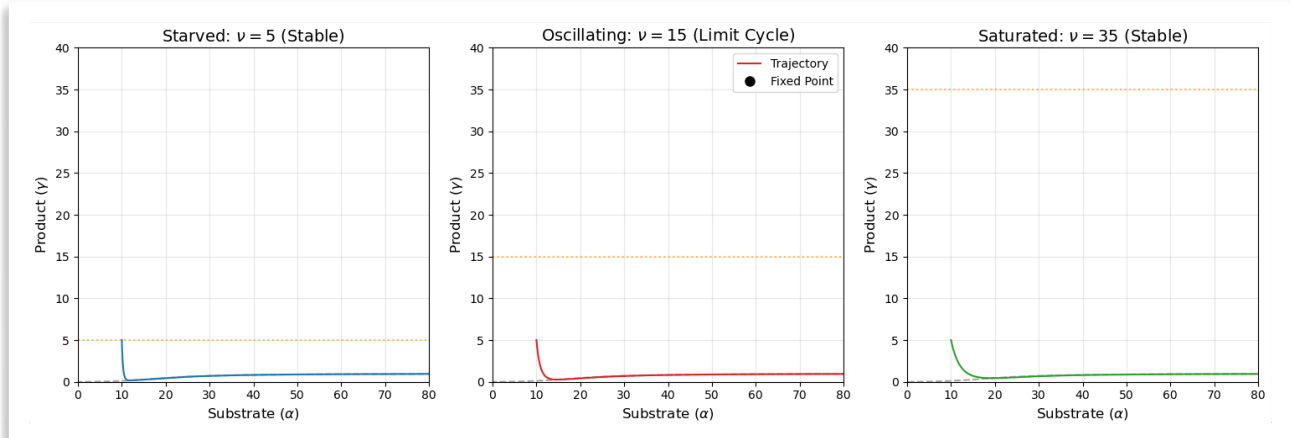


Figure 6.5: Phase Plane portraits across the three dynamic regimes. Izhikevich, E. M. (2007).

(Left) $\nu = 5$: Starved steady state. (Center) $\nu = 15$: Sustained relaxation oscillations. (Right) $\nu = 35$: Saturated steady state. The red dot represents the equilibrium point shifting along the S-shaped fast nullcline.

6.4.2 Effect of the Allosteric Constant (L)

The parameter L measures the allosteric conformation ratio of the phosphofructokinase (PFK) enzyme. It controls the degree of nonlinearity and cooperativity in the positive feedback loop.

The shape of the fast nullcline (the critical manifold M_0) is determined by L through the algebraic relation:

$$\alpha = \frac{k\gamma L + (1 + \gamma)^2}{\sigma (1 + \gamma)^2}$$

- High Cooperativity ($L \gg 1$): A large value of L (e.g., $L = 1000$ or higher) creates a deeply folded, pronounced S-shape. This ensures a broad unstable middle branch, accommodating a wide range of input rates ν that yield oscillations. The amplitude of the substrate $A\alpha$ becomes very large.
- Low Cooperativity ($L \rightarrow 0$): As L decreases, the "knees" of the fast nullcline unfold. Below a critical threshold L_c , the function loses its non-monotonicity, the fold points vanish, and the manifold becomes strictly monotonically increasing. Under these conditions, the unstable branch disappears ($\frac{\partial g}{\partial \gamma} < 0$ globally), and relaxation oscillations become mathematically impossible regardless of ν .

6.4.3 Sensitivity to the Perturbation Parameter (ϵ)

While ϵ is treated as an infinitesimally small constant in singular perturbation theory, numerical simulations require a finite value. The parameter $\epsilon = \frac{1}{q}$ explicitly controls the stiffness of the system and the speed of the fast transitions.

As $\epsilon \rightarrow 0$, the numerical trajectory aligns perfectly with the geometric boundaries of the critical manifold M_0 , and the time spent during the rapid jumps approaches zero. The total period approaches the theoretical asymptotic limit T_{slow} (Fenichel, 1979):

$$\lim_{\epsilon \rightarrow 0} T(\epsilon) = \int_{M_0^-} dt + \int_{M_0^+} dt$$

For larger, finite values of ϵ , the trajectories deviate from M_0 by an order of $O(\epsilon)$ due to Fenichel's theorem (Jones, 1995). The jumps are no longer perfectly horizontal, and the sharp corners of the limit cycle round off. If ϵ becomes $O(1)$, the system completely loses its fast-slow time-scale separation, the relaxation characteristic is destroyed, and the oscillations become simple, smooth harmonic cycles—or cease entirely if the fixed point regains stability.

6.4.4 Two-Parameter Bifurcation Space

Although one-dimensional bifurcation diagrams adeptly depict the shift from a steady state to a limit cycle with the variation of a single parameter, biological systems function within a multi-dimensional parameter space. To comprehensively delineate the resilience of the glycolytic oscillator, it is imperative to compute the bounds of the Hopf bifurcation across numerous concurrently variable parameters.

By methodically assessing the eigenvalues of the Jacobian matrix—particularly delineating the parameter range where the trace transitions from negative to positive values (the instability criterion)—we can formulate a two-parameter bifurcation diagram (Kuznetsov, 2004).

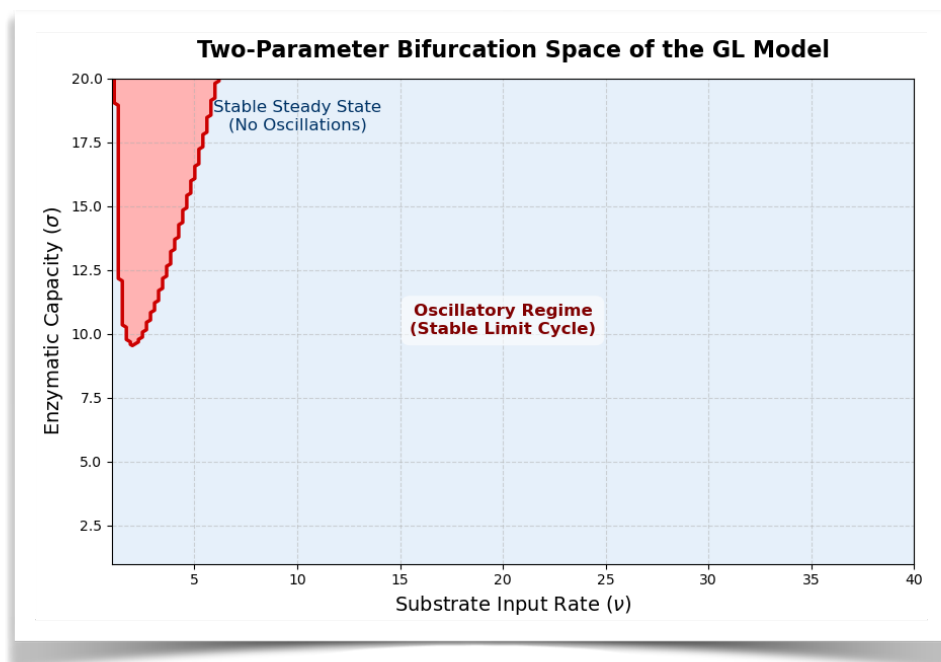


Figure 6.4: Two-Parameter Bifurcation Diagram mapping the oscillatory domain. Kuznetsov, Y. A. (2004).

The shaded region represents the parameter combinations of Substrate Input Rate (ν) and Enzymatic Capacity (σ) that yield sustained relaxation oscillations (where the steady state loses stability). Outside this bounded "island" of instability, the system naturally relaxes into a stable, non-oscillatory steady state.

This phase diagram is highly biologically significant. It demonstrates that the metabolic clock is not merely dependent on a specific glucose influx (ν), but is intimately tied to the total concentration and maximum velocity of the available PFK enzyme (σ). If the enzymatic capacity drops too low, oscillations cannot be sustained regardless of the substrate input, representing a strict boundary for metabolic homeostasis.

Chapter 7: Discussion

7.1 Biological Interpretation of the Mathematical Results

The mathematical examination of the Goldbeter-Lefever (GL) model through geometric singular perturbation theory offers significant insights into cellular control mechanisms. The abstract mathematical constructs we developed—namely the S-shaped critical manifold, Hopf bifurcations, and relaxation oscillations—possess concrete physical implications in the glycolytic pathway.

1. **The S-Shaped Manifold as a Biochemical Switch** In Chapter 5, we showed that the critical manifold M_0 forms a folded, S-shaped curve due to the high allosteric constant L and the cooperativity of the enzyme. Biologically, this represents phosphofructokinase (PFK) functioning as a highly sensitive biochemical switch. The lower branch (M_0^-) corresponds to the T-state (tense, inactive) dominated enzyme, where substrate is conserved. The upper branch (M_0^+) corresponds to the R-state (relaxed, active) enzyme, which rapidly consumes the substrate. The unstable middle branch represents a physiological impossibility: a threshold that the cell cannot sustainably occupy, forcing it to decisively jump between the "on" and "off" states.

2. **Fast-Slow Dynamics and Cellular Efficiency** The mathematical requirement for a small perturbation parameter ($\epsilon \ll 1$) strictly mirrors the physical reality of enzyme kinetics. The conformational changes of the PFK tetramer and the subsequent binding of ADP/AMP occur in milliseconds, while the bulk accumulation of fructose-6-phosphate (F6P) takes seconds to minutes. This time-scale separation allows the cell to "store" energy continuously and "release" it in explosive, efficient bursts, preventing the wasteful, continuous draining of metabolic pools that would occur in a simple, non-allosteric linear pathway.

3. **Bifurcations as Metabolic Boundaries** The Hopf bifurcations identified at ν_{min} and ν_{max} represent the strict physiological boundaries of the cell's metabolic capacity. If the glucose infusion rate (ν) is too low, the cell starves, resting in a stable, low-energy steady state. If the infusion rate is too high, the enzyme becomes permanently saturated, and the system loses its rhythmic regulation, risking a toxic accumulation of metabolic byproducts. The oscillatory window between these bifurcation points represents the optimal homeostatic regime where the cell can adaptively respond to varying energy demands.

7.2 Comparison with Experimental Data

The ultimate test of any mathematical model in systems biology is its fidelity to in vitro and in vivo experimental observations. Despite its structural simplicity, the two-variable GL model exhibits remarkable qualitative and quantitative agreement with classical experimental data.

- **Waveform Morphology:** Experimental fluorometric measurements of NADH (a downstream proxy for PFK activity) in yeast extracts by Ghosh and Chance (1964) revealed distinct, pulse-like spikes interspersed with slow accumulation phases. This matches the asymmetric relaxation oscillation limit cycles predicted by our fast-slow phase plane analysis, confirming that the "bursting" mechanism is accurately captured by the equations.
- **Periodicity:** The GL model, when scaled with realistic physiological rate constants for yeast PFK, naturally produces oscillation periods on the order of 1 to 10 minutes. This

perfectly aligns with the macroscopic biological clock observed in actual yeast suspensions.

- **Response to Substrate Flow:** In experimental setups, slowly increasing the continuous infusion of glucose transitions the yeast extract from a quiescent state to an oscillating state, and finally to a saturated steady state. This sequence perfectly mirrors the theoretical bifurcation diagram (Section 6.4), validating the existence of the critical boundaries v_{min} and v_{max} .

7.3 Limitations of the Goldbeter-Lefever Model

While the GL model is a triumph of conceptual reductionism, it is an idealized abstraction. As a theoretical biologist, it is crucial to acknowledge the mathematical and physical limitations of this framework.

1. **Route Aggregation:** The GL model significantly condenses a complex ten-step biochemical route into merely two variables (substrate α and product γ). It disregards the particular intermediate kinetics of enzymes such as hexokinase, aldolase, and pyruvate kinase. Therefore, it fails to consider intricate downstream feedback mechanisms, such as the inhibition of hexokinase by glucose-6-phosphate or the feed-forward activation of pyruvate kinase.
2. **Presumption of a Homogeneously Mixed System:** The model comprises ordinary differential equations (ODEs), which basically presume that the cytoplasm is an ideally homogeneous region. Cells are intricately organised, and localised concentration gradients are present. A more precise, dimensionally correct model necessitates the use of partial differential equations (PDEs) to represent the reaction-diffusion dynamics of metabolites inside the viscous internal fluid.
3. **The MWC Concerted Assumption:** The derivation relies solely on the Monod-Wyman-Changeux (MWC) model (Monod, Wyman, & Changeux, 1965), which asserts that all four subunits of PFK simultaneously transition from the T state to the R state. Experimental structural biology has revealed that numerous allosteric enzymes exhibit sequential binding characteristics, better adequately depicted by the Koshland-Némethy-Filmer (KNF) model (Koshland, Némethy, & Filmer, 1966). The GL model does not consider these complex, intermediate hybrid states.
4. **Deterministic versus Stochastic Dynamics:** The equations are entirely deterministic. In individual, microscopic cells, the total quantity of enzyme molecules may be sufficiently low for thermodynamic noise and stochastic fluctuations to become relevant. At low copy numbers, the continuous limit cycle may deteriorate into unpredictable, noisy bursts that cannot be described by continuous differential equations.

7.4 The Impact of Stochastic Noise in Small Cellular Volumes

A fundamental limitation of utilizing continuous Ordinary Differential Equations (ODEs) to model intracellular kinetics is the inherent assumption of the thermodynamic limit. The Goldbeter-Lefever model mathematically assumes that the cellular cytoplasm contains a continuous, infinite pool of reacting molecules, thereby producing a perfectly deterministic limit cycle. However, physical reality dictates that a biological cell is a microscopic, confined volume containing a discrete and sometimes very small number of enzyme and metabolite molecules.

In such mesoscopic systems, the timing of individual molecular collisions and enzymatic reactions is governed by probability, leading to thermodynamic noise. When the total copy number of Phosphofructokinase (PFK) or ATP molecules is sufficiently low, deterministic stability analyses (such as the Jacobian evaluation) begin to break down. To capture this physical reality, the system must be analyzed using stochastic simulation algorithms, such as the exact Gillespie Stochastic Simulation Algorithm (SSA).

Under stochastic conditions, the strict geometric boundaries of the fast-slow phase plane are blurred. The trajectory does not traverse a perfect, infinitely repeatable loop. Instead, the system exhibits "quasi-cycles"—noisy, irregular bursts of enzymatic activity that roughly follow the critical manifold but exhibit significant variance in both amplitude and periodicity. While the deterministic GL model elegantly proves the existence of the oscillatory attractor, incorporating stochastic chemical kinetics is essential for understanding how the cellular clock maintains robustness against the inherent randomness of the molecular environment.

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